



Syllabus: Sets, Relations, Functions, Partial orders, Lattices, **Monoids**, Groups.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	1	1	0	1	1	0	1	0	1	0
2 Marks Count	0	0	1	1	1	1	2	0	2	1	0
Total Marks	0	1	3	2	3	3	4	1	4	3	0

Welcome to the "Discrete Mathematics: Set Theory & Algebra" chapter, a cornerstone of computer science that underpins many advanced topics. This section is crucial for the GATE Computer Science exam, typically accounting for **8-12 marks**. Questions often test your foundational understanding of definitions, properties, and the application of theorems, ranging from direct formula recall to multi-step problem-solving involving logical deduction. Mastering these concepts is not just about scoring well; it's about building the analytical and problem-solving skills essential for a career in computing.

Topic-wise Key Concepts

Set Theory

Set theory is the fundamental language for describing collections of objects. It provides the basis for understanding relations, functions, and many other discrete structures.

Definition and Core Idea:

A set is a well-defined collection of distinct objects, called elements. Sets are typically denoted by capital letters, and their elements are enclosed in curly braces. The core idea is to classify and manipulate collections of items.

Important Formulas and Results:

- **Cardinality:** The number of elements in a set A is denoted by $|A|$.
- **Power Set:** The set of all subsets of A , denoted $P(A)$. If $|A| = n$, then $|P(A)| = 2^n$.
- **Principle of Inclusion-Exclusion:**
 - For two sets: $|A \cup B| = |A| + |B| - |A \cap B|$
 - For three sets: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Key Properties and Identities:

- **Commutative Laws:**
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- **Associative Laws:**
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive Laws:**
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De Morgan's Laws:**
 - $\overline{A \cup B} = \overline{A} \cap \overline{B}$
 - $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- **Idempotent Laws:**
 - $A \cup A = A$
 - $A \cap A = A$
- **Identity Laws:**
 - $A \cup \emptyset = A$
 - $A \cap U = A$ (where U is the universal set)
- **Complement Laws:**

- $A \cup \overline{A} = U$
- $A \cap \overline{A} = \emptyset$
- $\overline{\emptyset} = U$
- $\overline{U} = \emptyset$
- $\overline{\overline{A}} = A$

Common Pitfalls & Tricky Points:

- Confusing elements with sets containing those elements (e.g., $a \in \{a\}$ but $a \neq \{a\}$).
- Misinterpreting the empty set $\emptyset$. Remember $\emptyset$ is a subset of every set, and $P(\emptyset) = \{\emptyset\}$.
- Errors in applying De Morgan's Laws or Inclusion-Exclusion for complex scenarios.

Problem-Solving Techniques:

- Use Venn diagrams for visualizing set operations and proving identities for up to three sets.
- For formal proofs, use element-wise arguments (e.g., to prove $A \subseteq B$, show that if $x \in A$, then $x \in B$).
- Apply the Principle of Inclusion-Exclusion systematically for counting problems involving overlapping sets.

Relations

A relation describes how elements from one set relate to elements of another set (or the same set). It's a fundamental concept for modeling connections and dependencies.

Definition and Core Idea:

A relation R from set A to set B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say a is related to b , often written aRb . The core idea is to define specific associations between elements.

Important Formulas and Results:

- **Number of relations:** If $|A| = m$ and $|B| = n$, the number of relations from A to B is 2^{mn} .
- **Types of Relations on a Set A :**
 1. **Reflexive:** For all $a \in A$, $(a, a) \in R$.
 2. **Irreflexive:** For all $a \in A$, $(a, a) \notin R$.
 3. **Symmetric:** For all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \in R$.
 4. **Antisymmetric:** For all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$.
 5. **Asymmetric:** For all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \notin R$. (Implies irreflexive and antisymmetric).
 6. **Transitive:** For all $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.
- **Equivalence Relation:** A relation that is Reflexive, Symmetric, and Transitive.
- **Equivalence Class:** If R is an equivalence relation on A , the equivalence class of an element $a \in A$ is $[a]_R = \{x \in A \mid (x, a) \in R\}$. Equivalence classes partition the set A .
- **Partial Order Relation:** A relation that is Reflexive, Antisymmetric, and Transitive. (See Partial Order section for more details).

Key Properties and Identities:

- The inverse of a relation R is $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.
- Composition of relations: If $R_1 : A \rightarrow B$ and $R_2 : B \rightarrow C$, then $R_2 \circ R_1 = \{(a, c) \mid \exists b \in B, (a, b) \in R_1 \text{ and } (b, c) \in R_2\}$.

Common Pitfalls & Tricky Points:

- Confusing symmetric with antisymmetric. A relation can be neither, both (only if it contains only pairs (a, a)), or one but not the other.
- Verifying transitivity requires checking all possible chains (a, b) and (b, c) .
- Forgetting to check reflexivity for equivalence or partial order relations.

Problem-Solving Techniques:

- Represent relations using directed graphs (digraphs) or adjacency matrices, especially for small sets.
- To prove a relation is of a certain type, systematically check each property by definition.
- For equivalence relations, identify the equivalence classes to understand the partition of the set.

Functions

Functions are special types of relations where each input maps to exactly one output. They are central to all areas of mathematics and computer science.

Definition and Core Idea:

A function f from set A (domain) to set B (codomain), denoted $f : A \rightarrow B$, assigns to each element $x \in A$ exactly one element $y \in B$. The core idea is a deterministic mapping from inputs to outputs.

Important Formulas and Results:

- **Range:** The set of all actual output values, $\text{range}(f) = \{f(x) \mid x \in A\}$. Note that $\text{range}(f) \subseteq B$.
- **Types of Functions:**
 1. **Injective (One-to-one):** If $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in A$. Each element in the codomain is mapped to by at most one element in the domain.
 2. **Surjective (Onto):** For every $y \in B$, there exists at least one $x \in A$ such that $f(x) = y$. The range of f is equal to its codomain B .
 3. **Bijective (One-to-one and Onto):** A function that is both injective and surjective.
- **Inverse Function:** A function $f : A \rightarrow B$ has an inverse function $f^{-1} : B \rightarrow A$ if and only if f is bijective. Then $f^{-1}(y) = x \iff f(x) = y$.
- **Composition of Functions:** If $f : A \rightarrow B$ and $g : B \rightarrow C$, then $(g \circ f) : A \rightarrow C$ is defined as $(g \circ f)(x) = g(f(x))$.
- **Number of Functions:** If $|A| = m$ and $|B| = n$:
 - Total functions from A to B : n^m
 - Injective functions from A to B : $P(n, m) = \frac{n!}{(n-m)!}$ if $m \leq n$, else 0.
 - Bijective functions from A to B : $n!$ if $m = n$, else 0.
 - Surjective functions from A to B : $n! \cdot S(m, n)$, where $S(m, n)$ is the Stirling number of the second kind. Alternatively, using Inclusion-Exclusion: $\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$.

Key Properties and Identities:

- Composition of functions is associative: $(h \circ g) \circ f = h \circ (g \circ f)$.
- If f is bijective, then $f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$, where I_A and I_B are identity functions on sets A and B respectively.

Common Pitfalls & Tricky Points:

- Confusing the codomain with the range.
- Incorrectly assuming a function has an inverse when it's not bijective.
- Proving surjectivity requires showing that every element in the codomain has a preimage, not just some.

Problem-Solving Techniques:

- To prove injectivity, assume $f(x_1) = f(x_2)$ and algebraically show $x_1 = x_2$.
- To prove surjectivity, take an arbitrary y from the codomain and show there exists an x in the domain such that $f(x) = y$.
- When counting functions, carefully identify the constraints (injective, surjective, bijective).

Binary Operation

A binary operation combines two elements from a set to produce a third element within the same set. It's the foundation

for algebraic structures like groups and rings.

Definition and Core Idea:

A binary operation $*$ on a set S is a function $*$: $S \times S \rightarrow S$. For any $a, b \in S$, $a * b$ is a unique element in S . The core idea is to define an internal way to combine elements.

Important Formulas and Results:

- **Properties of a Binary Operation:**

1. **Closure:** For all $a, b \in S$, $a * b \in S$. (This is inherent in the definition).
2. **Associativity:** For all $a, b, c \in S$, $(a * b) * c = a * (b * c)$.
3. **Commutativity:** For all $a, b \in S$, $a * b = b * a$.
4. **Identity Element:** An element $e \in S$ such that for all $a \in S$, $a * e = e * a = a$. If it exists, it is unique.
5. **Inverse Element:** For an element $a \in S$, an element $a^{-1} \in S$ is its inverse if $a * a^{-1} = a^{-1} * a = e$, where e is the identity element. If an identity exists, inverses are unique for each element.

Key Properties and Identities:

- If an identity element exists, it is unique.
- If an identity element exists and the operation is associative, then inverses are unique.

Common Pitfalls & Tricky Points:

- Forgetting to check closure: the result of the operation must always be in the original set.
- Misidentifying the identity element or assuming it always exists.
- Assuming commutativity or associativity when not explicitly stated or proven.

Problem-Solving Techniques:

- To check properties, test with arbitrary elements or counterexamples.
- For finite sets, construct a Cayley table (operation table) to check properties visually.

Group Theory

Group theory studies algebraic structures called groups, which are sets equipped with a binary operation satisfying specific axioms. It's fundamental in cryptography, coding theory, and physics.

Definition and Core Idea:

A group $(G, *)$ is a set G together with a binary operation $*$ that satisfies four axioms:

1. **Closure:** For all $a, b \in G$, $a * b \in G$.
2. **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **Identity Element:** There exists an element $e \in G$ such that for all $a \in G$, $a * e = e * a = a$.
4. **Inverse Element:** For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

The core idea is to formalize symmetry and transformations.

Important Formulas and Results:

- **Related Structures:**

- **Semigroup:** A set with an associative binary operation.
- **Monoid:** A semigroup with an identity element.
- **Abelian Group (Commutative Group):** A group where the operation is commutative ($a * b = b * a$).
- **Subgroup:** A non-empty subset H of a group G is a subgroup if H is itself a group under the same operation.
 - **Subgroup Test:** H is a subgroup of G if for all $a, b \in H$, $a * b^{-1} \in H$.
- **Cyclic Group:** A group G is cyclic if there exists an element $g \in G$ such that every element of G can be written as a power of g (i.e., $G = \{g^n \mid n \in \mathbb{Z}\}$). g is called a generator.
- **Order of an element:** The smallest positive integer n such that $a^n = e$. If no such n exists, the element has infinite order.

order.

- **Lagrange's Theorem:** If G is a finite group and H is a subgroup of G , then the order of H divides the order of G (i.e., $|H|$ divides $|G|$).

Key Properties and Identities:

- The identity element is unique.
- Every element has a unique inverse.
- $(a^{-1})^{-1} = a$
- $(a * b)^{-1} = b^{-1} * a^{-1}$

Common Pitfalls & Tricky Points:

- Forgetting to check all four group axioms, especially the existence of inverses for *every* element.
- Confusing the order of an element with the order of the group.
- Misapplying Lagrange's Theorem (it states a necessary condition for a subgroup, not a sufficient one).

Problem-Solving Techniques:

- To prove a set with an operation is a group, systematically verify all four axioms.
- To find subgroups, look for subsets that are closed under the operation and contain the identity and inverses.
- Use Lagrange's Theorem to quickly rule out potential subgroup orders.

Identity Function

The identity function is a fundamental concept in functions and algebraic structures, acting as a neutral element for function composition.

Definition and Core Idea:

For any set A , the identity function on A , denoted I_A or id_A , is a function $I_A : A \rightarrow A$ such that for every element $x \in A$, $I_A(x) = x$. The core idea is a mapping that leaves elements unchanged.

Important Formulas and Results:

- If $f : A \rightarrow B$ is any function, then $f \circ I_A = f$ and $I_B \circ f = f$.
- If $f : A \rightarrow B$ is a bijective function, then its inverse $f^{-1} : B \rightarrow A$ satisfies $f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$.

Key Properties and Identities:

- The identity function is always bijective (one-to-one and onto).
- It serves as the identity element for the operation of function composition.

Common Pitfalls & Tricky Points:

- Confusing the identity function with a constant function (e.g., $f(x) = c$).
- Forgetting that the identity function is specific to a set A .

Problem-Solving Techniques:

- Use the properties of the identity function when simplifying compositions or proving inverse relationships.

Lattice

Lattices are special types of partially ordered sets where every pair of elements has a unique "least upper bound" and "greatest lower bound." They are crucial in areas like logic, set theory, and computer science for modeling hierarchical structures.

Definition and Core Idea:

A poset $(L, \preceq)$ is a lattice if for every pair of elements $a, b \in L$, both their least upper bound (LUB or join, denoted $a \vee b$) and greatest lower bound (GLB or meet, denoted $a \wedge b$) exist and are unique. The core idea is a structure where elements always have well-defined "closest" common superiors and inferiors.

Important Formulas and Results:

- **Join (Supremum):** $a \vee b = \sup\{a, b\}$.
- **Meet (Infimum):** $a \wedge b = \inf\{a, b\}$.
- **Types of Lattices:**
 - **Distributive Lattice:** A lattice where the distributive laws hold:
 - $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
 - $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
 - **Complemented Lattice:** A bounded lattice (has a greatest element 1 and a least element 0) where every element a has a complement a' such that $a \wedge a' = 0$ and $a \vee a' = 1$.
 - **Boolean Algebra:** A distributive and complemented lattice.

Key Properties and Identities:

- **Idempotent Laws:** $a \wedge a = a, a \vee a = a$
- **Commutative Laws:** $a \wedge b = b \wedge a, a \vee b = b \vee a$
- **Associative Laws:** $(a \wedge b) \wedge c = a \wedge (b \wedge c), (a \vee b) \vee c = a \vee (b \vee c)$
- **Absorption Laws:** $a \wedge (a \vee b) = a, a \vee (a \wedge b) = a$
- **Duality Principle:** If a statement is true for a lattice, its dual (exchanging $\wedge$ with $\vee$, $\preceq$ with $\succeq$, 0 with 1) is also true.

Common Pitfalls & Tricky Points:

- Not every poset is a lattice. A poset fails to be a lattice if any pair of elements lacks a unique LUB or GLB.
- Distinguishing between maximal/minimal elements and greatest/least elements in a poset (a lattice always has a greatest and least element if it's bounded).

Problem-Solving Techniques:

- Draw Hasse diagrams to visually identify LUBs and GLBs for all pairs.
- To check for distributivity, look for sublattices isomorphic to N_5 (pentagon lattice) or M_3 (diamond lattice) in the Hasse diagram (they are non-distributive).

Mathematical Induction

Mathematical Induction is a powerful proof technique used to establish the truth of a statement for all natural numbers (or a subset thereof). It's essential for proving properties of algorithms, data structures, and number theory results.

Definition and Core Idea:

The Principle of Mathematical Induction states that to prove a statement $P(n)$ is true for all integers $n \geq n_0$:

1. **Base Case:** Show that $P(n_0)$ is true.
2. **Inductive Hypothesis:** Assume that $P(k)$ is true for some arbitrary integer $k \geq n_0$.
3. **Inductive Step:** Show that $P(k+1)$ is true, using the inductive hypothesis.

The core idea is like a chain reaction: if the first domino falls, and falling dominoes always knock over the next one, then all dominoes will fall.

Important Formulas and Results:

- **Principle of Strong Induction:**
 1. **Base Case:** Show that $P(n_0)$ is true.
 2. **Inductive Hypothesis:** Assume that $P(j)$ is true for all integers j such that $n_0 \leq j \leq k$, for some arbitrary integer $k \geq n_0$.
 3. **Inductive Step:** Show that $P(k+1)$ is true, using the inductive hypothesis.

Key Properties and Identities:

- Both standard and strong induction are equivalent in power; strong induction is often more convenient for proofs where $P(k+1)$ depends on more than just $P(k)$.

Common Pitfalls & Tricky Points:

- Incorrect base case: The statement must hold for the starting value.
- Assuming what needs to be proven in the inductive step. The inductive hypothesis must be explicitly used.
- Not clearly stating the inductive hypothesis or the inductive step.
- Algebraic errors during the inductive step.

Problem-Solving Techniques:

- Clearly label each step: Base Case, Inductive Hypothesis, Inductive Step.
- In the inductive step, try to manipulate the expression for $P(k+1)$ to isolate an instance of $P(k)$ that can be substituted using the hypothesis.
- For inequalities, sometimes it's easier to prove $P(k+1) - P(k) \geq 0$ or similar.

Number Theory

Number theory is the study of integers and their properties, including divisibility, prime numbers, and modular arithmetic. It has direct applications in cryptography, error correction codes, and algorithm analysis.

Definition and Core Idea:

Number theory explores the relationships and properties of integers. The core idea is to understand the structure and behavior of numbers under various operations.

Important Formulas and Results:

- **Divisibility:** $a|b$ means a divides b (i.e., $b = ka$ for some integer k).
- **Division Algorithm:** For integers a (dividend) and d (divisor) with $d > 0$, there exist unique integers q (quotient) and r (remainder) such that $a = dq + r$, where $0 \leq r < d$.
- **Greatest Common Divisor (GCD):** $\gcd(a, b)$ is the largest integer that divides both a and b .
 - **Euclidean Algorithm:** An efficient method to compute GCD. $\gcd(a, b) = \gcd(b, a \bmod b)$.
 - **Bézout's Identity:** For any integers a, b , there exist integers x, y such that $ax + by = \gcd(a, b)$.
- **Least Common Multiple (LCM):** $\text{lcm}(a, b)$ is the smallest positive integer divisible by both a and b .
 - Relationship: $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$.
- **Prime Numbers:** A positive integer greater than 1 with exactly two positive divisors: 1 and itself.
 - **Fundamental Theorem of Arithmetic:** Every integer greater than 1 can be uniquely represented as a product of prime numbers (up to the order of factors).
- **Modular Arithmetic:** $a \equiv b \pmod{m}$ means $m|(a-b)$. This is an equivalence relation.
 - Properties:
 - If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.
 - $(a \bmod m) + (b \bmod m) \bmod m = (a + b) \bmod m$
 - $(a \bmod m) \cdot (b \bmod m) \bmod m = (a \cdot b) \bmod m$
- **Fermat's Little Theorem:** If p is a prime number and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$. Also, $a^p \equiv a \pmod{p}$ for any integer a .
- **Euler's Totient Function $\phi(n)$:** Counts the number of positive integers up to n that are relatively prime to n .
 - If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ is the prime factorization of n , then $\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right)$.
 - If p is prime, $\phi(p) = p - 1$.
- **Euler's Theorem:** If n is a positive integer and a is an integer relatively prime to n (i.e., $\gcd(a, n) = 1$), then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Key Properties and Identities:

- Congruence modulo m is an equivalence relation.

- The set of integers modulo n , $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$, forms a ring under addition and multiplication modulo n . If n is prime, $\mathbb{Z}_n$ is a field.

Common Pitfalls & Tricky Points:

- Mistakes in applying modular arithmetic, especially with negative numbers or division. Remember division is not always well-defined in modular arithmetic (requires modular inverse).
- Confusing prime numbers with relatively prime numbers.
- Incorrectly calculating $\phi(n)$ for composite numbers.

Problem-Solving Techniques:

- Use the Euclidean algorithm for GCD and extended Euclidean algorithm for modular inverses.
- Simplify large powers in modular arithmetic using Fermat's Little Theorem or Euler's Theorem.
- For divisibility problems, use properties of congruences.

Onto (Surjective Functions)

Onto functions, also known as surjective functions, ensure that every element in the codomain is "hit" by at least one element from the domain. This property is vital for understanding mappings that cover their entire target set.

Definition and Core Idea:

A function $f : A \rightarrow B$ is said to be onto (or surjective) if for every element $y \in B$ (the codomain), there exists at least one element $x \in A$ (the domain) such that $f(x) = y$. The core idea is that the function's range completely fills its codomain.

Important Formulas and Results:

- A function $f : A \rightarrow B$ is surjective if and only if $\text{range}(f) = B$.
- If A and B are finite sets, and $f : A \rightarrow B$ is surjective, then $|A| \geq |B|$.
- The number of surjective functions from a set of size m to a set of size n is given by $\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$.

Key Properties and Identities:

- If f and g are surjective, then $g \circ f$ is surjective.
- If $g \circ f$ is surjective, then g must be surjective (but f need not be).

Common Pitfalls & Tricky Points:

- Confusing surjectivity with injectivity. An onto function doesn't require unique preimages.
- Failing to demonstrate a preimage for an arbitrary element in the codomain, instead just showing some elements have preimages.

Problem-Solving Techniques:

- To prove a function $f : A \rightarrow B$ is surjective, pick an arbitrary $y \in B$, then show how to construct an $x \in A$ such that $f(x) = y$. This often involves solving $y = f(x)$ for x .
- For finite sets, compare the cardinalities of the domain and codomain. If $|A| < |B|$, no surjective function from A to B can exist.

Partial Order

Partial order relations define a sense of relative order among elements in a set, but not necessarily for every pair. They are crucial for modeling hierarchies, dependencies, and precedence in computer science (e.g., task scheduling, class inheritance).

Definition and Core Idea:

A relation R on a set A is a partial order relation if it is reflexive, antisymmetric, and transitive. A set A with a partial order relation $\preceq$ is called a partially ordered set (poset), denoted $(A, \preceq)$. The core idea is to establish an ordering where some elements might be incomparable.

Important Formulas and Results:

- **Properties of Partial Order:**

1. **Reflexive:** For all $a \in A$, $a \preceq a$.
2. **Antisymmetric:** For all $a, b \in A$, if $a \preceq b$ and $b \preceq a$, then $a = b$.
3. **Transitive:** For all $a, b, c \in A$, if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

- **Comparability:** Two elements $a, b \in A$ are comparable if $a \preceq b$ or $b \preceq a$. Otherwise, they are incomparable.

- **Total Order (Linear Order):** A partial order where every pair of elements is comparable.

- **Hasse Diagram:** A graphical representation of a finite poset, where:

- Elements are represented by nodes.
- An edge goes upwards from a to b if $a \preceq b$ and there is no c such that $a \preceq c \preceq b$ (i.e., b covers a).
- Reflexive loops and transitive edges are omitted.

- **Elements in a Poset:**

- **Maximal Element:** An element a such that there is no $b \in A$ with $a \preceq b$ and $a \neq b$.
- **Minimal Element:** An element a such that there is no $b \in A$ with $b \preceq a$ and $a \neq b$.
- **Greatest Element (Maximum):** An element g such that for all $x \in A$, $x \preceq g$. (If it exists, it is unique).
- **Least Element (Minimum):** An element l such that for all $x \in A$, $l \preceq x$. (If it exists, it is unique).
- **Upper Bound:** For a subset $S \subseteq A$, an element $u \in A$ is an upper bound of S if $s \preceq u$ for all $s \in S$.
- **Lower Bound:** For a subset $S \subseteq A$, an element $l \in A$ is a lower bound of S if $l \preceq s$ for all $s \in S$.
- **Least Upper Bound (LUB / Supremum):** The least element among all upper bounds of S , denoted $\sup(S)$.
- **Greatest Lower Bound (GLB / Infimum):** The greatest element among all lower bounds of S , denoted $\inf(S)$.

Key Properties and Identities:

- A greatest element is always a maximal element, but a maximal element is not necessarily the greatest. Similarly for least and minimal elements.
- A finite poset always has at least one maximal and one minimal element.

Common Pitfalls & Tricky Points:

- Confusing maximal/minimal elements with greatest/least elements. A poset can have multiple maximal elements but at most one greatest element.
- Errors in drawing Hasse diagrams, especially omitting necessary edges or including redundant ones.
- Incorrectly identifying LUBs or GLBs for subsets.

Problem-Solving Techniques:

- To verify a partial order, systematically check reflexivity, antisymmetry, and transitivity.
- Draw Hasse diagrams to visualize the order and identify special elements (maximal, minimal, greatest, least, LUB, GLB).
- For LUB/GLB, first find all upper/lower bounds, then identify the least/greatest among them.

Polynomials

Polynomials are fundamental algebraic expressions used extensively in computer science for modeling, interpolation, error correction, and algorithm analysis (e.g., complexity). In discrete mathematics, they often appear in contexts like finite fields and generating functions.

Definition and Core Idea:

A polynomial in a variable x is an expression of the form $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where a_i are coefficients (often real or complex numbers, but can be from finite fields in discrete math) and n is a non-negative integer called the degree of the polynomial (if $a_n \neq 0$). The core idea is to represent relationships using sums of powers of a variable.

Important Formulas and Results:

- **Degree:** The highest power of x with a non-zero coefficient.
- **Root (or Zero):** A value c such that $P(c) = 0$.
- **Polynomial Division Algorithm:** For polynomials $P(x)$ (dividend) and $D(x)$ (divisor) with $D(x) \neq 0$, there exist unique polynomials $Q(x)$ (quotient) and $R(x)$ (remainder) such that $P(x) = D(x)Q(x) + R(x)$, where $\deg(R(x)) < \deg(D(x))$ or $R(x) = 0$.
- **Remainder Theorem:** If a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$.
- **Factor Theorem:** $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$ (i.e., c is a root of $P(x)$).
- **Fundamental Theorem of Algebra:** Every non-constant polynomial with complex coefficients has at least one complex root. Consequently, a polynomial of degree n has exactly n complex roots (counting multiplicity).
- **Vieta's Formulas:** Relate the coefficients of a polynomial to the sums and products of its roots. For a quadratic $ax^2 + bx + c = 0$ with roots r_1, r_2 : $r_1 + r_2 = -b/a$, $r_1 r_2 = c/a$.

Key Properties and Identities:

- A polynomial of degree n has at most n distinct roots.
- Polynomials form a ring under addition and multiplication.

Common Pitfalls & Tricky Points:

- Algebraic errors in polynomial division or evaluating polynomials.
- Forgetting to count root multiplicities when applying the Fundamental Theorem of Algebra.
- Not considering the domain of coefficients (e.g., real vs. complex vs. finite field).

Problem-Solving Techniques:

- Use synthetic division for quick division by $(x - c)$.
- Apply the Remainder and Factor Theorems to find roots or factors efficiently.
- For problems involving roots, Vieta's formulas can often provide shortcuts.

Quick Formula Reference

- **Set Theory:**
 - $|P(A)| = 2^{|A|}$
 - $|A \cup B| = |A| + |B| - |A \cap B|$
 - De Morgan's Laws: $\overline{A \cup B} = \overline{A} \cap \overline{B}$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- **Relations:**
 - Number of relations from A to B : $2^{|A||B|}$
 - Equivalence Relation: Reflexive, Symmetric, Transitive.
 - Partial Order: Reflexive, Antisymmetric, Transitive.
- **Functions (from $|A| = m$ to $|B| = n$):**
 - Total functions: n^m
 - Injective functions: $P(n, m) = \frac{n!}{(n-m)!}$ (if $m \leq n$)
 - Bijective functions: $n!$ (if $m = n$)
 - Surjective functions: $\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$
 - Identity function $I_A(x) = x$
- **Group Theory:**
 - Group Axioms: Closure, Associativity, Identity, Inverse.
 - Lagrange's Theorem: $|H|$ divides $|G|$ for subgroup H of group G .
- **Lattice:**
 - Poset where every pair has unique LUB (join $\vee$) and GLB (meet $\wedge$).
 - Distributive Lattice: $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$, etc.
- **Mathematical Induction:**
 - Base Case $P(n_0)$, Inductive Hypothesis $P(k)$, Inductive Step $P(k+1)$.
- **Number Theory:**
 - Division Algorithm: $a = dq + r$, $0 \leq r < d$
 - $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$
 - Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod{p}$ (for prime p , $p \nmid a$)

- Euler's Totient Function: $\phi(n) = n \prod_{p|n, p \text{ prime}} \left(1 - \frac{1}{p}\right)$
- Euler's Theorem: $a^{\phi(n)} \equiv 1 \pmod{n}$ (for $\gcd(a, n) = 1$)
- **Polynomials:**
 - Remainder Theorem: $P(x) \pmod{(x - c)} = P(c)$
 - Factor Theorem: $(x - c)$ is a factor $\iff P(c) = 0$

Important Tips for GATE

1. **Master Definitions and Properties:** Many GATE questions directly test your understanding of definitions (e.g., what makes a relation symmetric?) and properties (e.g., De Morgan's laws). Memorize them thoroughly.
2. **Practice with Hasse Diagrams:** For partial orders and lattices, drawing Hasse diagrams is crucial. Practice identifying maximal/minimal elements, greatest/least elements, and LUB/GLB from diagrams.
3. **Systematic Verification:** When asked to check if a structure is a group, a relation is an equivalence relation, or a function is bijective, systematically go through each axiom/property. Don't skip steps or assume properties.
4. **Number Theory Shortcuts:** Familiarize yourself with the Euclidean algorithm for GCD, modular arithmetic properties, and theorems like Fermat's Little Theorem and Euler's Theorem. These are vital for efficiently solving problems involving large numbers or powers.
5. **Inclusion-Exclusion Principle:** This principle is frequently tested for counting problems in set theory and functions (especially surjective functions). Practice applying it for 2, 3, and general cases.
6. **Mathematical Induction Structure:** Always clearly state your base case, inductive hypothesis, and inductive step. Ensure your inductive step explicitly uses the inductive hypothesis to avoid circular reasoning.
7. **Cardinality vs. Countability:** Understand the difference between finite, countably infinite, and uncountable sets. Be familiar with standard examples (e.g., $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable; $\mathbb{R}, P(\mathbb{N})$ are uncountable) and Cantor's diagonalization argument.
8. **Watch for "If and Only If":** Pay close attention to conditions that are "if and only if" (iff) as they imply equivalence and are often key to proving properties or identifying specific structures.

4.1

Binary Operation (8)

4.1.1 Binary Operation: GATE CSE 1989 | Question: 1-v



The number of possible commutative binary operations that can be defined on a set of n elements (for a given n) is _____.

gate1989 descriptive set-theory&algebra binary-operation

Answer key

4.1.2 Binary Operation: GATE CSE 1994 | Question: 2.2



On the set N of non-negative integers, the binary operation _____ is associative and non-commutative.

gate1994 set-theory&algebra normal group-theory binary-operation fill-in-the-blanks

Answer key

4.1.3 Binary Operation: GATE CSE 2003 | Question: 38



Consider the set $\{a, b, c\}$ with binary operators $+$ and $*$ defined as follows:

$+$	a	b	c
a	b	a	c
b	a	b	c
c	a	c	b

$*$	a	b	c
a	a	b	c
b	b	c	a
c	c	c	b

For example, $a + c = c$, $c + a = a$, $c * b = c$ and $b * c = a$.

Given the following set of equations:

- $(a * x) + (a * y) = c$
- $(b * x) + (c * y) = c$

The number of solution(s) (i.e., pair(s) (x, y) that satisfy the equations) is

A. 0

B. 1

C. 2

D. 3

gatecse-2003 set-theory&algebra normal binary-operation

Answer key

4.1.4 Binary Operation: GATE CSE 2006 | Question: 28



A logical binary relation $\odot$, is defined as follows:

A	B	$A \odot B$
True	True	True
True	False	True
False	True	False
False	False	True

Let $\sim$ be the unary negation (NOT) operator, with higher precedence than $\odot$.

Which one of the following is equivalent to $A \wedge B$?

A. $(\sim A \odot B)$ B. $\sim (A \odot \sim B)$ C. $\sim (\sim A \odot \sim B)$ D. $\sim (\sim A \odot B)$

gatecse-2006 set-theory&algebra binary-operation propositional-logic mathematical-logic

Answer key

4.1.5 Binary Operation: GATE CSE 2013 | Question: 1



A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

A. Commutative but not associative

B. Both commutative and associative

C. Associative but not commutative

D. Neither commutative nor associative

gatecse-2013 set-theory&algebra easy binary-operation

Answer key

4.1.6 Binary Operation: GATE CSE 2015 | Set 1 | Question: 28



The binary operator $\neq$ is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\neq$?

A. Both commutative and associative

B. Commutative but not associative

C. Not commutative but associative

D. Neither commutative nor associative

gatecse-2015-set1 set-theory&algebra easy binary-operation boolean-algebra

Answer key

4.1.7 Binary Operation: GATE CSE 2015 | Set 3 | Question: 2



Let $\#$ be the binary operator defined as

$X \# Y = X' + Y'$ where X and Y are Boolean variables.

Consider the following two statements.

- $(S_1) (P \# Q) \# R = P \# (Q \# R)$

• $(S_2) Q \# R = (R \# Q)$

Which are the following is/are true for the Boolean variables P, Q and R ?

- A. Only S_1 is true
 C. Both S_1 and S_2 are true
 B. Only S_2 is true
 D. Neither S_1 nor S_2 are true

gatecse-2015-set3 set-theory&algebra binary-operation normal

Answer key

4.1.8 Binary Operation: GATE IT 2006 | Question: 2



For the set N of natural numbers and a binary operation $f : N \times N \rightarrow N$, an element $z \in N$ is called an identity for f , if $f(a, z) = a = f(z, a)$, for all $a \in N$. Which of the following binary operations have an identity?

- I. $f(x, y) = x + y - 3$
 II. $f(x, y) = \max(x, y)$
 III. $f(x, y) = x^y$

- A. I and II only
 B. II and III only
 C. I and III only
 D. None of these

gateit-2006 set-theory&algebra easy binary-operation

Answer key

4.2 Countable Uncountable Set (2)

4.2.1 Countable Uncountable Set: GATE CSE 1994 | Question: 3.9



Every subset of a countable set is countable.

State whether the above statement is true or false with reason.

gate1994 set-theory&algebra normal set-theory countable-uncountable-set true-false

Answer key

4.2.2 Countable Uncountable Set: GATE CSE 2018 | Question: 27



Let N be the set of natural numbers. Consider the following sets,

- P : Set of Rational numbers (positive and negative)
- Q : Set of functions from $\{0, 1\}$ to N
- R : Set of functions from N to $\{0, 1\}$
- S : Set of finite subsets of N

Which of the above sets are countable?

- A. Q and S only
 B. P and S only
 C. P and R only
 D. P, Q and S only

gatecse-2018 set-theory&algebra countable-uncountable-set normal two-marks

Answer key

4.3 Functions (30)

4.3.1 Functions: GATE CSE 1987 | Question: 9b



How many one-to-one functions are there from a set A with n elements onto itself?

gate1987 set-theory&algebra functions descriptive

Answer key

4.3.2 Functions: GATE CSE 1988 | Question: 13ii



If the set S has a finite number of elements, prove that if f maps S onto S , then f is one-to-one.

Answer key

4.3.3 Functions: GATE CSE 1989 | Question: 13c



Find the number of single valued functions from set A to another set B , given that the cardinalities of the sets A and B are m and n respectively.

Answer key

4.3.4 Functions: GATE CSE 1993 | Question: 8.6



Let A and B be sets with cardinalities m and n respectively. The number of one-one mappings from A to B , when $m < n$, is

- A. m^n B. nP_m C. mC_n D. nC_m
 E. mP_n

Answer key

4.3.5 Functions: GATE CSE 1996 | Question: 1.3



Suppose X and Y are sets and $|X|$ and $|Y|$ are their respective cardinality. It is given that there are exactly 97 functions from X to Y . From this one can conclude that

- A. $|X| = 1, |Y| = 97$ B. $|X| = 97, |Y| = 1$
 C. $|X| = 97, |Y| = 97$ D. None of the above

Answer key

4.3.6 Functions: GATE CSE 1996 | Question: 2.1



Let R denote the set of real numbers. Let $f: R \times R \rightarrow R \times R$ be a bijective function defined by $f(x, y) = (x + y, x - y)$. The inverse function of f is given by

- A. $f^{-1}(x, y) = \left(\frac{1}{x+y}, \frac{1}{x-y}\right)$ B. $f^{-1}(x, y) = (x - y, x + y)$
 C. $f^{-1}(x, y) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$ $f^{-1}(x, y) = [2(x - y), 2(x + y)]$

Answer key

4.3.7 Functions: GATE CSE 1997 | Question: 13



Let F be the set of one-to-one functions from the set $\{1, 2, \dots, n\}$ to the set $\{1, 2, \dots, m\}$ where $m \geq n \geq 1$.

- How many functions are members of F ?
- How many functions f in F satisfy the property $f(i) = 1$ for some $i, 1 \leq i \leq n$?
- How many functions f in F satisfy the property $f(i) < f(j)$ for all $i, j, 1 \leq i \leq j \leq n$?

Answer key

4.3.8 Functions: GATE CSE 1998 | Question: 1.8



The number of functions from an m element set to an n element set is

- A. $m + n$ B. m^n C. n^m D. $m * n$

gate1998 set-theory&algebra combinatory functions easy

Answer key

4.3.9 Functions: GATE CSE 2001 | Question: 2.3



Let $f : A \rightarrow B$ a function, and let E and F be subsets of A . Consider the following statements about images.

- $S_1 : f(E \cup F) = f(E) \cup f(F)$
- $S_2 : f(E \cap F) = f(E) \cap f(F)$

Which of the following is true about S_1 and S_2 ?

- A. Only S_1 is correct B. Only S_2 is correct
C. Both S_1 and S_2 are correct D. None of S_1 and S_2 is correct

gatecse-2001 set-theory&algebra functions normal

Answer key

4.3.10 Functions: GATE CSE 2001 | Question: 4



Consider the function $h : N \times N \rightarrow N$ so that $h(a, b) = (2a + 1)2^b - 1$, where $N = \{0, 1, 2, 3, \dots\}$ is the set of natural numbers.

- a. Prove that the function h is an injection (one-one).
b. Prove that it is also a Surjection (onto)

gatecse-2001 functions set-theory&algebra normal descriptive

Answer key

4.3.11 Functions: GATE CSE 2003 | Question: 37



Let $f : A \rightarrow B$ be an injective (one-to-one) function. Define $g : 2^A \rightarrow 2^B$ as:

$g(C) = \{f(x) \mid x \in C\}$, for all subsets C of A .

Define $h : 2^B \rightarrow 2^A$ as: $h(D) = \{x \mid x \in A, f(x) \in D\}$, for all subsets D of B . Which of the following statements is always true?

- A. $g(h(D)) \subseteq D$ B. $g(h(D)) \supseteq D$
C. $g(h(D)) \cap D = \phi$ D. $g(h(D)) \cap (B - D) \neq \phi$

gatecse-2003 set-theory&algebra functions difficult

Answer key

4.3.12 Functions: GATE CSE 2003 | Question: 39



Let $\Sigma = \{a, b, c, d, e\}$ be an alphabet. We define an encoding scheme as follows:

$g(a) = 3, g(b) = 5, g(c) = 7, g(d) = 9, g(e) = 11$.

Let p_i denote the i -th prime number ($p_1 = 2$).

For a non-empty string $s = a_1 \dots a_n$, where each $a_i \in \Sigma$, define $f(s) = \prod_{i=1}^n P_i^{g(a_i)}$.

For a non-empty sequence $\langle s_1, \dots, s_n \rangle$ of strings from Σ^+ , define $h(\langle s_1 \dots s_n \rangle) = \prod_{i=1}^n P_i^{f(s_i)}$

Which of the following numbers is the encoding, h , of a non-empty sequence of strings?

- A. $2^7 3^7 5^7$ B. $2^8 3^8 5^8$ C. $2^9 3^9 5^9$ D. $2^{10} 3^{10} 5^{10}$

gatecse-2003 set-theory&algebra functions difficult

Answer key

4.3.13 Functions: GATE CSE 2005 | Question: 43



Let $f : B \rightarrow C$ and $g : A \rightarrow B$ be two functions and let $h = fog$. Given that h is an onto function which one of the following is TRUE?

- A. f and g should both be onto functions
- B. f should be onto but g need not to be onto
- C. g should be onto but f need not be onto
- D. both f and g need not be onto

gatecse-2005 set-theory&algebra functions normal

Answer key

4.3.14 Functions: GATE CSE 2006 | Question: 2



Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subsets of W . The number of functions from Z to E is

- A. $z^{2^{xy}}$
- B. $z \times 2^{xy}$
- C. $z^{2^{x+y}}$
- D. 2^{xyz}

gatecse-2006 set-theory&algebra normal functions

Answer key

4.3.15 Functions: GATE CSE 2006 | Question: 25



Let $S = \{1, 2, 3, \dots, m\}, m > 3$. Let $X_1, \dots, X_n$ be subsets of S each of size 3. Define a function f from S to the set of natural numbers as, $f(i)$ is the number of sets X_j that contain the element i . That is $f(i) = |\{j \mid i \in X_j\}|$ then $\sum_{i=1}^m f(i)$ is:

- A. $3m$
- B. $3n$
- C. $2m + 1$
- D. $2n + 1$

gatecse-2006 set-theory&algebra normal functions

Answer key

4.3.16 Functions: GATE CSE 2007 | Question: 3



What is the maximum number of different Boolean functions involving n Boolean variables?

- A. n^2
- B. 2^n
- C. 2^{2^n}
- D. 2^{n^2}

gatecse-2007 combinatory functions normal

Answer key

4.3.17 Functions: GATE CSE 2012 | Question: 37



How many onto (or surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

- A. 2^n
- B. $2^n - 1$
- C. $2^n - 2$
- D. $2(2^n - 2)$

gatecse-2012 set-theory&algebra functions normal

Answer key

4.3.18 Functions: GATE CSE 2014 | Set 1 | Question: 50



Let S denote the set of all functions $f : \{0, 1\}^4 \rightarrow \{0, 1\}$. Denote by N the number of functions from S to the set $\{0, 1\}$. The value of $\log_2 \log_2 N$ is _____.

gatecse-2014-set1 set-theory&algebra functions combinatory numerical-answers

Answer key

4.3.19 Functions: GATE CSE 2014 | Set 3 | Question: 2



Let X and Y be finite sets and $f : X \rightarrow Y$ be a function. Which one of the following statements is TRUE?

- A. For any subsets A and B of X , $|f(A \cup B)| = |f(A)| + |f(B)|$
- B. For any subsets A and B of X , $f(A \cap B) = f(A) \cap f(B)$
- C. For any subsets A and B of X , $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$
- D. For any subsets S and T of Y , $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.20 Functions: GATE CSE 2014 | Set 3 | Question: 49



Consider the set of all functions $f : \{0, 1, \dots, 2014\} \rightarrow \{0, 1, \dots, 2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$. Consider the following statements:

- P . For each such function it must be the case that for every i , $f(i) = i$.
- Q . For each such function it must be the case that for some i , $f(i) = i$.
- R . Each function must be onto.

Which one of the following is CORRECT?

- A. P, Q and R are true
- B. Only Q and R are true
- C. Only P and Q are true
- D. Only R is true

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.21 Functions: GATE CSE 2015 | Set 1 | Question: 5



If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

- A. $\frac{h(x)}{g(x)}$
- B. $\frac{-1}{x}$
- C. $\frac{g(x)}{h(x)}$
- D. $\frac{x}{(1-x)^2}$

gatecse-2015-set1 set-theory&algebra functions normal

Answer key

4.3.22 Functions: GATE CSE 2015 | Set 2 | Question: 40



The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is _____.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.23 Functions: GATE CSE 2015 | Set 2 | Question: 54



Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y . Let f be randomly chosen from F . The probability of f being one-to-one is _____.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.24 Functions: GATE CSE 2016 | Set 1 | Question: 28



A function $f : \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$f(n) = f(n/2) \text{ if } n \text{ is even}$$

$$f(n) = f(n+5) \text{ if } n \text{ is odd}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____.

gatecse-2016-set1 set-theory&algebra functions normal numerical-answers

Answer key

4.3.25 Functions: GATE CSE 2021 | Set 2 | Question: 11



Consider the following sets, where $n \geq 2$:

- S_1 : Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$
- S_2 : Set of all functions from the set $\{0, 1, 2, \dots, n^2 - 1\}$ to the set $\{0, 1, 2\}$

Which of the following choice(s) is/are correct?

- A. There does not exist a bijection from S_1 to S_2
- B. There exists a surjection from S_1 to S_2
- C. There exists a bijection from S_1 to S_2
- D. There does not exist an injection from S_1 to S_2

gatecse-2021-set2 multiple-selects set-theory&algebra functions one-mark

Answer key

4.3.26 Functions: GATE CSE 2023 | Question: 39



Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as

$$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2),$$

where $a_1, a_2 \in A$. Let $\mathcal{E} = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \mathcal{E} \rightarrow B$ as

$$F([x]) = f(x), \quad \text{for all the equivalence classes } [x] \text{ in } \mathcal{E}.$$

Which of the following statements is/are TRUE?

- A. F is NOT well-defined.
- B. F is an onto (or surjective) function.
- C. F is a one-to-one (or injective) function.
- D. F is a bijective function.

gatecse-2023 set-theory&algebra functions multiple-selects two-marks

Answer key

4.3.27 Functions: GATE CSE 2024 | Set 1 | Question: 22



Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is _____.

gatecse-2024-set1 numerical-answers set-theory&algebra functions one-mark

Answer key

4.3.28 Functions: GATE DA 2026 | Question: 23



The number of bijections $f(\cdot)$ from the set $S = \{1, 2, 3, 4\}$ to itself such that $f(f(n)) = n$, for all $n \in S$, is _____. (Answer in integer)

gateda-2026 set-theory&algebra functions numerical-answers one-mark

Answer key

4.3.29 Functions: GATE IT 2005 | Question: 31



Let f be a function from a set A to a set B , g a function from B to C , and h a function from A to C , such that $h(a) = g(f(a))$ for all $a \in A$. Which of the following statements is always true for all such functions f and g ?

- A. g is onto $\implies h$ is onto
 B. h is onto $\implies f$ is onto
 C. h is onto $\implies g$ is onto
 D. h is onto $\implies f$ and g are onto

gateit-2005 set-theory&algebra functions normal

Answer key

4.3.30 Functions: GATE IT 2006 | Question: 6



Given a boolean function $f(x_1, x_2, \dots, x_n)$, which of the following equations is NOT true?

- A. $f(x_1, x_2, \dots, x_n) = x'_1 f(x_1, x_2, \dots, x_n) + x_1 f(x_1, x_2, \dots, x_n)$
 B. $f(x_1, x_2, \dots, x_n) = x_2 f(x_1, x_2, \dots, x_n) + x'_2 f(x_1, x_2, \dots, x_n)$
 C. $f(x_1, x_2, \dots, x_n) = x'_n f(x_1, x_2, \dots, 0) + x_n f(x_1, x_2, \dots, 1)$
 D. $f(x_1, x_2, \dots, x_n) = f(0, x_2, \dots, x_n) + f(1, x_2, \dots, x_n)$

gateit-2006 set-theory&algebra functions normal

Answer key

4.4 Group Theory (33)

4.4.1 Group Theory: GATE CSE 1988 | Question: 2xviii



Show that if G is a group such that $(a * b)^2 = a^2 * b^2$ for all a, b belonging to G , then G is an abelian.

gate1988 descriptive group-theory

Answer key

4.4.2 Group Theory: GATE CSE 1990 | Question: 2-x



Match the pairs in the following questions:

(a) Groups	(p) Associativity
(b) Semigroups	(q) Identity
(c) Monoids	(r) Commutativity
(d) Abelian groups	(s) Left inverse

gate1990 match-the-following set-theory&algebra group-theory easy

Answer key

4.4.3 Group Theory: GATE CSE 1992 | Question: 14a



If G is a group of even order, then show that there exists an element $a \neq e$, the identity in G , such that $a^2 = e$.

gate1992 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.4 Group Theory: GATE CSE 1993 | Question: 28



Let $(\{p, q\}, *)$ be a semigroup where $p * p = q$. Show that:

- a. $p * q = q * p$ and
 b. $q * q = q$

Answer key

4.4.5 Group Theory: GATE CSE 1994 | Question: 1.10



Some group (G, o) is known to be abelian. Then, which one of the following is true for G ?

- A. $g = g^{-1}$ for every $g \in G$
- B. $g = g^2$ for every $g \in G$
- C. $(goh)^2 = g^2oh^2$ for every $g, h \in G$
- D. G is of finite order

gate1994 set-theory&algebra group-theory normal

Answer key

4.4.6 Group Theory: GATE CSE 1995 | Question: 2.17



Let A be the set of all non-singular matrices over real number and let $*$ be the matrix multiplication operation. Then

- A. A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup.
- B. $\langle A, * \rangle$ is a semigroup but not a monoid.
- C. $\langle A, * \rangle$ is a monoid but not a group.
- D. $\langle A, * \rangle$ is a group but not an abelian group.

gate1995 set-theory&algebra group-theory

Answer key

4.4.7 Group Theory: GATE CSE 1995 | Question: 21



Let G_1 and G_2 be subgroups of a group G .

- A. Show that $G_1 \cap G_2$ is also a subgroup of G .
- B. Is $G_1 \cup G_2$ always a subgroup of G ?

gate1995 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.8 Group Theory: GATE CSE 1996 | Question: 1.4



Which of the following statements is FALSE?

- A. The set of rational numbers is an abelian group under addition
- B. The set of integers in an abelian group under addition
- C. The set of rational numbers form an abelian group under multiplication
- D. The set of real numbers excluding zero is an abelian group under multiplication

gate1996 set-theory&algebra group-theory normal

Answer key

4.4.9 Group Theory: GATE CSE 1996 | Question: 2.4



Which one of the following is false?

- A. The set of all bijective functions on a finite set forms a group under function composition
- B. The set $\{1, 2, \dots, p-1\}$ forms a group under multiplication mod p , where p is a prime number
- C. The set of all strings over a finite alphabet forms a group under concatenation
- D. A subset $S \neq \emptyset$ of G is a subgroup of the group $\langle G, * \rangle$ if and only if for any pair of elements $a, b \in S, a * b^{-1} \in S$

gate1996 set-theory&algebra normal set-theory group-theory

Answer key

4.4.10 Group Theory: GATE CSE 1997 | Question: 3.1



Let $(\mathbb{Z}, *)$ be an algebraic structure where $\mathbb{Z}$ is the set of integers and the operation $*$ is defined by $n * m = \max(n, m)$. Which of the following statements is true for $(\mathbb{Z}, *)$?

- A. $(\mathbb{Z}, *)$ is a monoid
- B. $(\mathbb{Z}, *)$ is an Abelian group
- C. $(\mathbb{Z}, *)$ is a group
- D. None of the above

gate1997 set-theory&algebra group-theory normal

Answer key

4.4.11 Group Theory: GATE CSE 1998 | Question: 12



Let $(A, *)$ be a semigroup. Furthermore, for every a and b in A , if $a \neq b$, then $a * b \neq b * a$.

- a. Show that for every a in A , $a * a = a$
- b. Show that for every a, b in A , $a * b * a = a$
- c. Show that for every a, b, c in A , $a * b * c = a * c$

gate1998 set-theory&algebra group-theory descriptive

Answer key

4.4.12 Group Theory: GATE CSE 1999 | Question: 4



Let G be a finite group and H be a subgroup of G . For $a \in G$, define $aH = \{ah \mid h \in H\}$.

- a. Show that $|aH| = |bH|$.
- b. Show that for every pair of elements $a, b \in G$, either $aH = bH$ or aH and bH are disjoint.
- c. Use the above to argue that the order of H must divide the order of G .

gate1999 set-theory&algebra group-theory descriptive proof

Answer key

4.4.13 Group Theory: GATE CSE 2000 | Question: 4



Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $\otimes$ denote multiplication modulo 8, that is, $x \otimes y = (xy) \bmod 8$

- a. Prove that $(\{0, 1\}, \otimes)$ is not a group.
- b. Write three distinct groups $(G, \otimes)$ where $G \subset S$ and G has 2 elements.

gatecse-2000 set-theory&algebra descriptive group-theory

Answer key

4.4.14 Group Theory: GATE CSE 2002 | Question: 1.6



Which of the following is true?

- A. The set of all rational negative numbers forms a group under multiplication.
- B. The set of all non-singular matrices forms a group under multiplication.
- C. The set of all matrices forms a group under multiplication.
- D. Both B and C are true.

gatecse-2002 set-theory&algebra group-theory normal

Answer key

4.4.15 Group Theory: GATE CSE 2003 | Question: 7



Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

- A. does not form a group
- B. forms a non-commutative group
- C. does not have a right identity element
- D. forms a group if the empty string is removed from Σ^*

gatecse-2003 set-theory&algebra group-theory normal

Answer key

4.4.16 Group Theory: GATE CSE 2004 | Question: 72



The following is the incomplete operation table of a 4—element group.

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

The last row of the table is

- A. $caeb$
- B. $cbae$
- C. $cbea$
- D. $ceab$

gatecse-2004 set-theory&algebra group-theory normal

Answer key

4.4.17 Group Theory: GATE CSE 2005 | Question: 13



The set $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a group under multiplication modulo 15. The inverses of 4 and 7 are respectively:

- A. 3 and 13
- B. 2 and 11
- C. 4 and 13
- D. 8 and 14

gatecse-2005 set-theory&algebra normal group-theory

Answer key

4.4.18 Group Theory: GATE CSE 2005 | Question: 46



Consider the set H of all 3×3 matrices of the type

$$\begin{pmatrix} a & f & e \\ 0 & b & d \\ 0 & 0 & c \end{pmatrix}$$

where a, b, c, d, e and f are real numbers and $abc \neq 0$. Under the matrix multiplication operation, the set H is:

- A. a group
- B. a monoid but not a group
- C. a semi group but not a monoid
- D. neither a group nor a semi group

gatecse-2005 set-theory&algebra group-theory normal

Answer key

4.4.19 Group Theory: GATE CSE 2006 | Question: 3



The set $\{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. Given below are four possible reasons. Which one of them is false?

- A. It is not closed
- B. 2 does not have an inverse
- C. 3 does not have an inverse
- D. 8 does not have an inverse

gatecse-2006 set-theory&algebra group-theory normal

Answer key

4.4.20 Group Theory: GATE CSE 2007 | Question: 21



How many different non-isomorphic Abelian groups of order 4 are there?

- A. 2 B. 3 C. 4 D. 5

gatecse-2007 group-theory normal

Answer key

4.4.21 Group Theory: GATE CSE 2009 | Question: 1



Which one of the following is **NOT** necessarily a property of a Group?

- A. Commutativity B. Associativity
C. Existence of inverse for every element D. Existence of identity

gatecse-2009 set-theory&algebra easy group-theory

Answer key

4.4.22 Group Theory: GATE CSE 2009 | Question: 22



For the composition table of a cyclic group shown below:

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

Which one of the following choices is correct?

- A. a, b are generators B. b, c are generators
C. c, d are generators D. d, a are generators

gatecse-2009 set-theory&algebra normal group-theory

Answer key

4.4.23 Group Theory: GATE CSE 2010 | Question: 4



Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms

- A. A Group B. A Ring C. An integral domain D. A field

gatecse-2010 set-theory&algebra normal group-theory

Answer key

4.4.24 Group Theory: GATE CSE 2014 | Set 3 | Question: 3



Let G be a group with 15 elements. Let L be a subgroup of G . It is known that $L \neq G$ and that the size of L is at least 4. The size of L is _____.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key

4.4.25 Group Theory: GATE CSE 2014 | Set 3 | Question: 50



There are two elements x, y in a group $(G, *)$ such that every element in the group can be written as a product of some number of x 's and y 's in some order. It is known that

$$x * x = y * y = x * y * x * y = y * x * y * x = e$$

where e is the identity element. The maximum number of elements in such a group is ____.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key

4.4.26 Group Theory: GATE CSE 2018 | Question: 19

Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is ____

gatecse-2018 group-theory numerical-answers set-theory&algebra one-mark

Answer key

4.4.27 Group Theory: GATE CSE 2019 | Question: 10

Let G be an arbitrary group. Consider the following relations on G :

- $R_1 : \forall a, b \in G, aR_1b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$
- $R_2 : \forall a, b \in G, aR_2b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relations?

- A. R_1 and R_2 B. R_1 only C. R_2 only D. Neither R_1 nor R_2

gatecse-2019 engineering-mathematics discrete-mathematics set-theory&algebra group-theory one-mark

Answer key

4.4.28 Group Theory: GATE CSE 2020 | Question: 18

Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is ____.

gatecse-2020 numerical-answers group-theory easy one-mark

Answer key

4.4.29 Group Theory: GATE CSE 2021 | Set 1 | Question: 34

Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?

- A. Both G and H are always cyclic B. G may not be cyclic, but H is always cyclic
- C. G is always cyclic, but H may not be cyclic D. Both G and H may not be cyclic

gatecse-2021-set1 set-theory&algebra group-theory two-marks

Answer key

4.4.30 Group Theory: GATE CSE 2022 | Question: 17

Which of the following statements is/are TRUE for a group G ?

- A. If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
- B. If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G .
- C. If the order of G is 2, then G is commutative.
- D. If G is commutative, then a subgroup of G need not be commutative.

gatecse-2022 set-theory&algebra group-theory multiple-selects one-mark

Answer key

4.4.31 Group Theory: GATE CSE 2023 | Question: 41

Let X be a set and 2^X denote the powerset of X .

Define a binary operation Δ on 2^X as follows:

$$A \Delta B = (A - B) \cup (B - A).$$

Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?

- A. H is a group.
- B. Every element in H has an inverse, but H is NOT a group.
- C. For every $A \in 2^X$, the inverse of A is the complement of A .
- D. For every $A \in 2^X$, the inverse of A is A .

gatecse-2023 set-theory&algebra group-theory multiple-selects two-marks

Answer key

4.4.32 Group Theory: GATE CSE 2024 | Set 1 | Question: 42

Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

- A. Operator $\diamond$ obeys the associative law
- B. Operator $\square$ obeys the associative law
- C. Operator $\diamond$ over the operator $\square$ obeys the distributive law
- D. Operator $\square$ over the operator $\diamond$ obeys the distributive law

gatecse-2024-set1 multiple-selects set-theory&algebra group-theory two-marks

Answer key

4.4.33 Group Theory: GATE CSE 2024 | Set 2 | Question: 53

Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____.

gatecse-2024-set2 numerical-answers set-theory&algebra group-theory two-marks

Answer key

4.5 Identify Function (1)

4.5.1 Identify Function: GATE CSE 2025 | Set 1 | Question: 39

$A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$, we define

$$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$

for every number n in A . Which of the following is/are CORRECT about the mathematical structure $(F, \odot)$?

- A. $(F, \odot)$ is an Abelian group.
- B. $(F, \odot)$ is an Abelian monoid.
- C. $(F, \odot)$ is a non-Abelian group.
- D. $(F, \odot)$ is a non-Abelian monoid.

gatecse2025-set1 set-theory&algebra identify-function group-theory multiple-selects easy two-marks

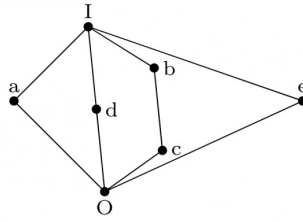
Answer key

4.6 Lattice (10)

4.6.1 Lattice: GATE CSE 1988 | Question: 1vii

The complement(s) of the element ' a ' in the lattice shown in below figure is (are) _____





gate1988 descriptive lattice set-theory&algebra

Answer key

4.6.2 Lattice: GATE CSE 1990 | Question: 17c



Show that the elements of the lattice $(N, \leq)$, where N is the set of positive integers and $a \leq b$ if and only if a divides b , satisfy the distributive property.

gate1990 descriptive set-theory&algebra lattice

Answer key

4.6.3 Lattice: GATE CSE 1992 | Question: 14b



Consider the set of integers $\{1, 2, 3, 4, 6, 8, 12, 24\}$ together with the two binary operations LCM (lowest common multiple) and GCD (greatest common divisor). Which of the following algebraic structures does this represent?

- A. group B. ring C. field D. lattice

gate1992 set-theory&algebra partial-order lattice normal

Answer key

4.6.4 Lattice: GATE CSE 1994 | Question: 2.9



The Hasse diagrams of all the lattices with up to four elements are _____ (write all the relevant Hasse diagrams)

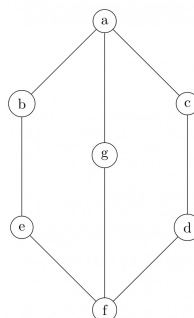
gate1994 set-theory&algebra lattice normal fill-in-the-blanks

Answer key

4.6.5 Lattice: GATE CSE 1997 | Question: 3.3



In the lattice defined by the Hasse diagram given in following figure, how many complements does the element 'e' have?



- A. 2 B. 3 C. 0 D. 1

gate1997 set-theory&algebra lattice normal

Answer key

4.6.6 Lattice: GATE CSE 2002 | Question: 4



$S = \{(1, 2), (2, 1)\}$ is binary relation on set $A = \{1, 2, 3\}$. Is it irreflexive? Add the minimum number of ordered pairs to S to make it an equivalence relation. Give the modified S .

Let $S = \{a, b\}$ and let $\square(S)$ be the powerset of S . Consider the binary relation ' $\subseteq$ ' (set inclusion)' on $\square(S)$. Draw the Hasse diagram corresponding to the lattice $(\square(S), \subseteq)$

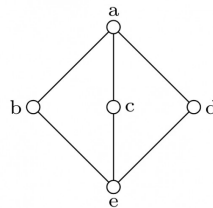
gatecse-2002 set-theory&algebra normal lattice descriptive

Answer key

4.6.7 Lattice: GATE CSE 2005 | Question: 9



The following is the Hasse diagram of the poset $[\{a, b, c, d, e\}, \prec]$



The poset is :

- A. not a lattice
- B. a lattice but not a distributive lattice
- C. a distributive lattice but not a Boolean algebra
- D. a Boolean algebra

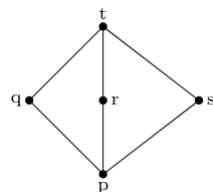
gatecse-2005 set-theory&algebra lattice normal

Answer key

4.6.8 Lattice: GATE CSE 2015 | Set 1 | Question: 34



Suppose $L = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:



For any $x, y \in L$, not necessarily distinct, $x \vee y$ and $x \wedge y$ are join and meet of x, y , respectively. Let $L^3 = \{(x, y, z) : x, y, z \in L\}$ be the set of all ordered triplets of the elements of L . Let p_r be the probability that an element $(x, y, z) \in L^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

- A. $p_r = 0$
- B. $p_r = 1$
- C. $0 < p_r \leq \frac{1}{5}$
- D. $\frac{1}{5} < p_r < 1$

gatecse-2015-set1 set-theory&algebra normal lattice

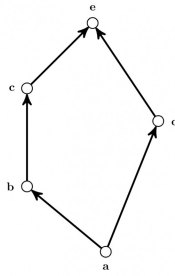
Answer key

4.6.9 Lattice: GATE CSE 2017 | Set 2 | Question: 21



Consider the set $X = \{a, b, c, d, e\}$ under partial ordering $R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}$

The Hasse diagram of the partial order (X, R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is _____

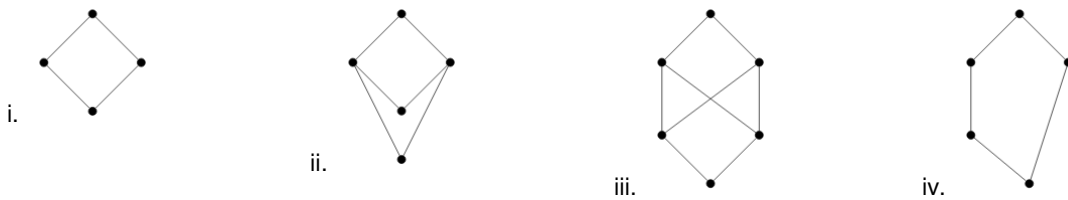
gatecse-2017-set2 set-theory&algebra lattice numerical-answers normal

Answer key

4.6.10 Lattice: GATE IT 2008 | Question: 28



Consider the following Hasse diagrams.



Which all of the above represent a lattice?

- A. (i) and (iv) only
 B. (ii) and (iii) only
 C. (iii) only
 D. (i), (ii) and (iv) only

gateit-2008 set-theory&algebra lattice normal

Answer key

4.7 Mathematical Induction (2)

4.7.1 Mathematical Induction: GATE CSE 1995 | Question: 23



Prove using mathematical induction for $n \geq 5, 2^n > n^2$

gate1995 set-theory&algebra proof mathematical-induction descriptive

Answer key

4.7.2 Mathematical Induction: GATE CSE 2000 | Question: 3



Consider the following sequence:

$s_1 = s_2 = 1$ and $s_i = 1 + \min(s_{i-1}, s_{i-2})$ for $i > 2$.

Prove by induction on n that $s_n = \lceil \frac{n}{2} \rceil$.

gatecse-2000 set-theory&algebra mathematical-induction descriptive

Answer key

4.8 Number Theory (7)

4.8.1 Number Theory: GATE CSE 1995 | Question: 7(A)



Determine the number of divisors of 600.

gate1995 set-theory&algebra number-theory numerical-answers

Answer key

4.8.2 Number Theory: GATE CSE 2014 | Set 2 | Question: 49



The number of distinct positive integral factors of 2014 is _____

gatecse-2014-set2 set-theory&algebra easy numerical-answers number-theory

Answer key

4.8.3 Number Theory: GATE CSE 2015 | Set 2 | Question: 9



The number of divisors of 2100 is _____.

gatecse-2015-set2 set-theory&algebra number-theory easy numerical-answers

Answer key

4.8.4 Number Theory: GATE IT 2005 | Question: 34



Let $n = p^2q$, where p and q are distinct prime numbers. How many numbers m satisfy $1 \leq m \leq n$ and $\gcd(m, n) = 1$? Note that $\gcd(m, n)$ is the greatest common divisor of m and n .

- A. $p(q-1)$ B. pq
C. $(p^2-1)(q-1)$ D. $p(p-1)(q-1)$

gateit-2005 set-theory&algebra normal number-theory

Answer key

4.8.5 Number Theory: GATE IT 2007 | Question: 16



The minimum positive integer p such that $3^p \pmod{17} = 1$ is

- A. 5 B. 8 C. 12 D. 16

gateit-2007 set-theory&algebra normal number-theory

Answer key

4.8.6 Number Theory: GATE IT 2008 | Question: 24



The exponent of 11 in the prime factorization of $300!$ is

- A. 27 B. 28 C. 29 D. 30

gateit-2008 set-theory&algebra normal number-theory

Answer key

4.8.7 Number Theory: GATE1991-15,a



Show that the product of the least common multiple and the greatest common divisor of two positive integers a and b is $a \times b$.

gate1991 set-theory&algebra normal number-theory proof descriptive

Answer key

4.9

Onto (1)

4.9.1 Onto: GATE CSE 2025 | Set 1 | Question: 7



$g(\cdot)$ is a function from A to B , $f(\cdot)$ is a function from B to C , and their composition defined as $f(g(\cdot))$ is a mapping from A to C .

If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$?

- A. $g(\cdot)$ must be an onto (surjective) function.
B. $g(\cdot)$ must be a one-to-one (injective) function.
C. $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
D. $g(\cdot)$ is not required to be a one-to-one or onto function.

Answer key

4.10

Partial Order (10)

4.10.1 Partial Order: GATE CSE 1991 | Question: 01,xiv



If the longest chain in a partial order is of length n , then the partial order can be written as a _____ of n antichains.

gate1991 set-theory&algebra partial-order normal fill-in-the-blanks

Answer key

4.10.2 Partial Order: GATE CSE 1993 | Question: 8.5



The less-than relation, $<$, on real number is

- A. a partial ordering since it is asymmetric and reflexive
- B. a partial ordering since it is antisymmetric and reflexive
- C. not a partial ordering because it is not asymmetric and not reflexive
- D. not a partial ordering because it is not antisymmetric and reflexive
- E. none of the above

gate1993 set-theory&algebra partial-order easy

Answer key

4.10.3 Partial Order: GATE CSE 1996 | Question: 1.2



Let $X = \{2, 3, 6, 12, 24\}$, Let $\leq$ be the partial order defined by $X \leq Y$ if x divides y . Number of edges in the Hasse diagram of $(X, \leq)$ is

- A. 3
- B. 4
- C. 9
- D. None of the above

gate1996 set-theory&algebra partial-order normal

Answer key

4.10.4 Partial Order: GATE CSE 1997 | Question: 6.1



A partial order $\leq$ is defined on the set $S = \{x, a_1, a_2, \dots, a_n, y\}$ as $x \leq_i a_i$ for all i and $a_i \leq y$ for all i , where $n \geq 1$. The number of total orders on the set S which contain the partial order $\leq$ is

- A. $n!$
- B. $n + 2$
- C. n
- D. 1

gate1997 set-theory&algebra partial-order normal

Answer key

4.10.5 Partial Order: GATE CSE 1998 | Question: 11



Suppose $A = \{a, b, c, d\}$ and Π_1 is the following partition of A

$$\Pi_1 = \{\{a, b, c\}, \{d\}\}$$

- a. List the ordered pairs of the equivalence relations induced by Π_1 .
- b. Draw the graph of the above equivalence relation.
- c. Let $\Pi_2 = \{\{a\}, \{b\}, \{c\}, \{d\}\}$

$$\Pi_3 = \{\{a, b, c, d\}\}$$

$$\text{and } \Pi_4 = \{\{a, b\}, \{c, d\}\}$$

Draw a Poset diagram of the poset, $\langle \{\Pi_1, \Pi_2, \Pi_3, \Pi_4\}, \text{refines} \rangle$.

gate1998 set-theory&algebra normal partial-order descriptive

Answer key

4.10.6 Partial Order: GATE CSE 2003 | Question: 31



Let $(S, \leq)$ be a partial order with two minimal elements a and b , and a maximum element c . Let $P: S \rightarrow \{\text{True}, \text{False}\}$ be a predicate defined on S . Suppose that $P(a) = \text{True}$, $P(b) = \text{False}$ and $P(x) \implies P(y)$ for all $x, y \in S$ satisfying $x \leq y$, where $\implies$ stands for logical implication. Which of the following statements CANNOT be true?

- A. $P(x) = \text{True}$ for all $x \in S$ such that $x \neq b$
- B. $P(x) = \text{False}$ for all $x \in S$ such that $x \neq a$ and $x \neq c$
- C. $P(x) = \text{False}$ for all $x \in S$ such that $b \leq x$ and $x \neq c$
- D. $P(x) = \text{False}$ for all $x \in S$ such that $a \leq x$ and $b \leq x$

gatecse-2003 set-theory&algebra partial-order normal propositional-logic

Answer key

4.10.7 Partial Order: GATE CSE 2004 | Question: 73



The inclusion of which of the following sets into

$$S = \{\{1, 2\}, \{1, 2, 3\}, \{1, 3, 5\}, \{1, 2, 4\}, \{1, 2, 3, 4, 5\}\}$$

is necessary and sufficient to make S a complete lattice under the partial order defined by set containment?

- A. $\{1\}$
- B. $\{1\}, \{2, 3\}$
- C. $\{1\}, \{1, 3\}$
- D. $\{1\}, \{1, 3\}, \{1, 2, 3, 4\}, \{1, 2, 3, 5\}$

gatecse-2004 set-theory&algebra partial-order normal

Answer key

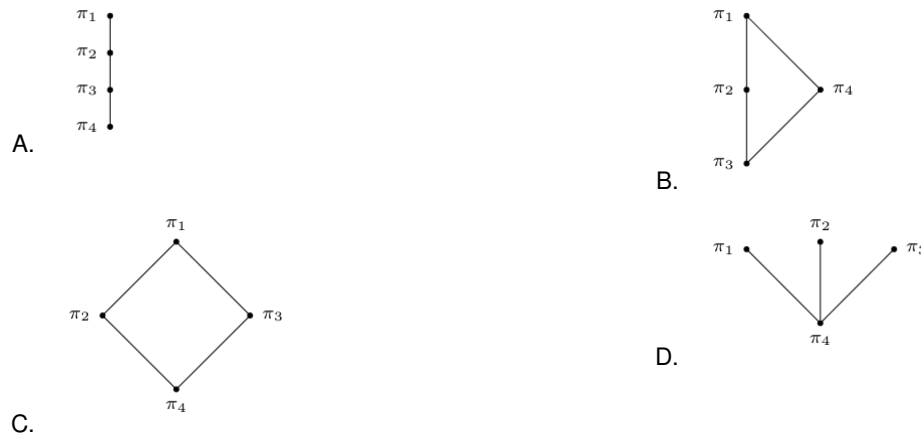
4.10.8 Partial Order: GATE CSE 2007 | Question: 26



Consider the set $S = \{a, b, c, d\}$. Consider the following 4 partitions $\pi_1, \pi_2, \pi_3, \pi_4$ on

$$S: \pi_1 = \{\overline{abcd}\}, \pi_2 = \{\overline{ab}, \overline{cd}\}, \pi_3 = \{\overline{abc}, \overline{d}\}, \pi_4 = \{\overline{a}, \overline{b}, \overline{c}, \overline{d}\}.$$

Let $\prec$ be the partial order on the set of partitions $S' = \{\pi_1, \pi_2, \pi_3, \pi_4\}$ defined as follows: $\pi_i \prec \pi_j$ if and only if π_i refines π_j . The poset diagram for $(S', \prec)$ is:



gatecse-2007 set-theory&algebra normal partial-order descriptive

Answer key

4.10.9 Partial Order: GATE CSE 2024 | Set 2 | Question: 24



Let P be the partial order defined on the set $\{1, 2, 3, 4\}$ as follows

$$P = \{(x, x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1, 2), (3, 2), (3, 4)\}$$

The number of total orders on $\{1, 2, 3, 4\}$ that contain P is _____.

Answer key

4.10.10 Partial Order: GATE IT 2007 | Question: 23



A partial order P is defined on the set of natural numbers as follows. Here $\frac{x}{y}$ denotes integer division.

- i. $(0, 0) \in P$.
- ii. $(a, b) \in P$ if and only if $(a \% 10) \leq (b \% 10)$ and $(\frac{a}{10}, \frac{b}{10}) \in P$.

Consider the following ordered pairs:

- i. $(101, 22)$
- ii. $(22, 101)$
- iii. $(145, 265)$
- iv. $(0, 153)$

Which of these ordered pairs of natural numbers are contained in P ?

- A. (i) and (iii)
- B. (ii) and (iv)
- C. (i) and (iv)
- D. (iii) and (iv)

gateit-2007 set-theory&algebra partial-order normal

Answer key

4.11 Polynomials (4)

4.11.1 Polynomials: GATE CSE 1997 | Question: 4.4



A polynomial $p(x)$ is such that $p(0) = 5, p(1) = 4, p(2) = 9$ and $p(3) = 20$. The minimum degree it should have is

- A. 1
- B. 2
- C. 3
- D. 4

gate1997 set-theory&algebra normal polynomials

Answer key

4.11.2 Polynomials: GATE CSE 2000 | Question: 2.4



A polynomial $p(x)$ satisfies the following:

- $p(1) = p(3) = p(5) = 1$
- $p(2) = p(4) = -1$

The minimum degree of such a polynomial is

- A. 1
- B. 2
- C. 3
- D. 4

gatecse-2000 set-theory&algebra normal polynomials

Answer key

4.11.3 Polynomials: GATE CSE 2014 | Set 2 | Question: 5



A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1, x = 2$ and $x = 3$. Which one of the following must be TRUE?

- A. $f(0)f(4) < 0$
- B. $f(0)f(4) > 0$
- C. $f(0) + f(4) > 0$
- D. $f(0) + f(4) < 0$

gatecse-2014-set2 set-theory&algebra polynomials normal

Answer key

4.11.4 Polynomials: GATE CSE 2017 | Set 2 | Question: 24



Consider the quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b . The solutions of this

equation in the same base b are $x = 5$ and $x = 6$. Then $b = \underline{\hspace{2cm}}$

gatecse-2017-set2 polynomials numerical-answers set-theory&algebra

Answer key

4.12

Relations (38)

4.12.1 Relations: GATE CSE 1987 | Question: 2d



State whether the following statements are TRUE or FALSE:

The union of two equivalence relations is also an equivalence relation.

gate1987 set-theory&algebra relations true-false

Answer key

4.12.2 Relations: GATE CSE 1987 | Question: 9a



How many binary relations are there on a set A with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key

4.12.3 Relations: GATE CSE 1987 | Question: 9e



How many true inclusion relations are there of the form $A \subseteq B$, where A and B are subsets of a set S with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key

4.12.4 Relations: GATE CSE 1989 | Question: 1-iv



The transitive closure of the relation $\{(1, 2), (2, 3), (3, 4), (5, 4)\}$ on the set $\{1, 2, 3, 4, 5\}$ is _____.

gate1989 set-theory&algebra relations descriptive

Answer key

4.12.5 Relations: GATE CSE 1992 | Question: 15.b



Let S be the set of all integers and let $n > 1$ be a fixed integer. Define for $a, b \in S$, aRb iff $a - b$ is a multiple of n . Show that R is an equivalence relation and find its equivalence classes for $n = 5$.

gate1992 set-theory&algebra normal relations descriptive

Answer key

4.12.6 Relations: GATE CSE 1994 | Question: 2.3



Amongst the properties {reflexivity, symmetry, anti-symmetry, transitivity} the relation $R = \{(x, y) \in N^2 | x \neq y\}$ satisfies _____

gate1994 set-theory&algebra normal relations fill-in-the-blanks

Answer key

4.12.7 Relations: GATE CSE 1995 | Question: 1.19



Let R be a symmetric and transitive relation on a set A . Then

- | | |
|---|---|
| A. R is reflexive and hence an equivalence relation | B. R is reflexive and hence a partial order |
| C. R is reflexive and hence not an equivalence relation | D. None of the above |

gate1995 set-theory&algebra relations normal

Answer key

4.12.8 Relations: GATE CSE 1996 | Question: 2.2



Let R be a non-empty relation on a collection of sets defined by $A R_B$ if and only if $A \cap B = \phi$. Then, (pick the true statement)

- A. A is reflexive and transitive
- B. R is symmetric and not transitive
- C. R is an equivalence relation
- D. R is not reflexive and not symmetric

gate1996 set-theory&algebra relations normal

Answer key

4.12.9 Relations: GATE CSE 1996 | Question: 8



Let F be the collection of all functions $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. If f and $g \in F$, define an equivalence relation $\sim$ by $f \sim g$ if and only if $f(3) = g(3)$.

- a. Find the number of equivalence classes defined by $\sim$.
- b. Find the number of elements in each equivalence class.

gate1996 set-theory&algebra relations functions normal descriptive

Answer key

4.12.10 Relations: GATE CSE 1997 | Question: 14



Let R be a reflexive and transitive relation on a set A . Define a new relation E on A as

$$E = \{(a, b) \mid (a, b) \in R \text{ and } (b, a) \in R\}$$

Prove that E is an equivalence relation on A .

Define a relation $\leq$ on the equivalence classes of E as $E_1 \leq E_2$ if $\exists a, b$ such that $a \in E_1, b \in E_2$ and $(a, b) \in R$. Prove that $\leq$ is a partial order.

gate1997 set-theory&algebra relations normal proof descriptive

Answer key

4.12.11 Relations: GATE CSE 1997 | Question: 6.3



The number of equivalence relations of the set $\{1, 2, 3, 4\}$ is

- A. 15
- B. 16
- C. 24
- D. 4

gate1997 set-theory&algebra relations normal group-theory

Answer key

4.12.12 Relations: GATE CSE 1998 | Question: 1.6



Suppose A is a finite set with n elements. The number of elements in the largest equivalence relation of A is

- A. n
- B. n^2
- C. 1
- D. $n + 1$

gate1998 set-theory&algebra relations easy

Answer key

4.12.13 Relations: GATE CSE 1998 | Question: 1.7



Let R_1 and R_2 be two equivalence relations on a set. Consider the following assertions:

- i. $R_1 \cup R_2$ is an equivalence relation
- ii. $R_1 \cap R_2$ is an equivalence relation

Which of the following is correct?

- A. Both assertions are true
- B. Assertion (i) is true but assertion (ii) is not true
- C. Assertion (ii) is true but assertion (i) is not true
- D. Neither (i) nor (ii) is true

gate1998 set-theory&algebra relations normal

Answer key

4.12.14 Relations: GATE CSE 1998 | Question: 10a

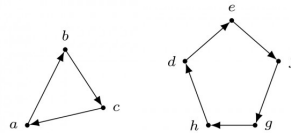
Prove by induction that the expression for the number of diagonals in a polygon of n sides is $\frac{n(n-3)}{2}$

gate1998 set-theory&algebra descriptive relations

Answer key

4.12.15 Relations: GATE CSE 1998 | Question: 10b

Let R be a binary relation on $A = \{a, b, c, d, e, f, g, h\}$ represented by the following two component digraph. Find the smallest integers m and n such that $m < n$ and $R^m = R^n$.



gate1998 descriptive set-theory&algebra relations

Answer key

4.12.16 Relations: GATE CSE 1998 | Question: 2.3

The binary relation $R = \{(1, 1), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$ on the set $A = \{1, 2, 3, 4\}$ is

- A. reflexive, symmetric and transitive
- B. neither reflexive, nor irreflexive but transitive
- C. irreflexive, symmetric and transitive
- D. irreflexive and antisymmetric

gate1998 set-theory&algebra easy relations

Answer key

4.12.17 Relations: GATE CSE 1999 | Question: 1.2

The number of binary relations on a set with n elements is:

- A. n^2
- B. 2^n
- C. 2^{n^2}
- D. None of the above

gate1999 set-theory&algebra relations combinatory easy

Answer key

4.12.18 Relations: GATE CSE 1999 | Question: 2.3

Let L be a set with a relation R which is transitive, anti-symmetric and reflexive and for any two elements $a, b \in L$, let the least upper bound $\text{lub}(a, b)$ and the greatest lower bound $\text{glb}(a, b)$ exist. Which of the following is/are true?

- A. L is a poset
- B. L is a Boolean algebra
- C. L is a lattice
- D. None of the above

gate1999 set-theory&algebra normal relations multiple-selects

Answer key

4.12.19 Relations: GATE CSE 1999 | Question: 3



- a. Mr. X claims the following:
If a relation R is both symmetric and transitive, then R is reflexive. For this, Mr. X offers the following proof:
“From xRy , using symmetry we get yRx . Now because R is transitive xRy and yRx together imply xRx . Therefore, R is reflexive”.
- b. Give an example of a relation R which is symmetric and transitive but not reflexive.

gate1999 set-theory&algebra relations normal descriptive

Answer key

4.12.20 Relations: GATE CSE 2000 | Question: 2.5



A relation R is defined on the set of integers as xRy iff $(x + y)$ is even. Which of the following statements is true?

- A. R is not an equivalence relation
- B. R is an equivalence relation having 1 equivalence class
- C. R is an equivalence relation having 2 equivalence classes
- D. R is an equivalence relation having 3 equivalence classes

gatecse-2000 set-theory&algebra relations normal

Answer key

4.12.21 Relations: GATE CSE 2001 | Question: 1.2



Consider the following relations:

- $R_1(a, b)$ iff $(a + b)$ is even over the set of integers
- $R_2(a, b)$ iff $(a + b)$ is odd over the set of integers
- $R_3(a, b)$ iff $a \cdot b > 0$ over the set of non-zero rational numbers
- $R_4(a, b)$ iff $|a - b| \leq 2$ over the set of natural numbers

Which of the following statements is correct?

- A. R_1 and R_2 are equivalence relations, R_3 and R_4 are not
- B. R_1 and R_3 are equivalence relations, R_2 and R_4 are not
- C. R_1 and R_4 are equivalence relations, R_2 and R_3 are not
- D. R_1, R_2, R_3 and R_4 all are equivalence relations

gatecse-2001 set-theory&algebra normal relations

Answer key

4.12.22 Relations: GATE CSE 2002 | Question: 2.17



The binary relation $S = \phi(\text{empty set})$ on a set $A = \{1, 2, 3\}$ is

- A. Neither reflexive nor symmetric
- B. Symmetric and reflexive
- C. Transitive and reflexive
- D. Transitive and symmetric

gatecse-2002 set-theory&algebra normal relations

Answer key

4.12.23 Relations: GATE CSE 2002 | Question: 3



Let A be a set of $n (> 0)$ elements. Let N_r be the number of binary relations on A and let N_f be the number of functions from A to A

- A. Give the expression for N_r , in terms of n .
- B. Give the expression for N_f , terms of n .

C. Which is larger for all possible n , N_r or N_f

gatecse-2002 set-theory&algebra normal descriptive relations

Answer key

4.12.24 Relations: GATE CSE 2004 | Question: 24



Consider the binary relation:

$$S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2\}\}$$

The reflexive transitive closure of S is

- A. $\{(x, y) \mid y > x \text{ and } x, y \in \{0, 1, 2\}\}$
- B. $\{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2\}\}$
- C. $\{(x, y) \mid y < x \text{ and } x, y \in \{0, 1, 2\}\}$
- D. $\{(x, y) \mid y \leq x \text{ and } x, y \in \{0, 1, 2\}\}$

gatecse-2004 set-theory&algebra easy relations

Answer key

4.12.25 Relations: GATE CSE 2005 | Question: 42



Let R and S be any two equivalence relations on a non-empty set A . Which one of the following statements is TRUE?

- A. $R \cup S, R \cap S$ are both equivalence relations
- B. $R \cup S$ is an equivalence relation
- C. $R \cap S$ is an equivalence relation
- D. Neither $R \cup S$ nor $R \cap S$ are equivalence relations

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.26 Relations: GATE CSE 2005 | Question: 7



The time complexity of computing the transitive closure of a binary relation on a set of n elements is known to be:

- A. $O(n)$
- B. $O(n \log n)$
- C. $O\left(n^{\frac{3}{2}}\right)$
- D. $O(n^3)$

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.27 Relations: GATE CSE 2006 | Question: 4



A relation R is defined on ordered pairs of integers as follows:

$$(x, y)R(u, v) \text{ if } x < u \text{ and } y > v$$

Then R is:

- A. Neither a Partial Order nor an Equivalence Relation
- B. A Partial Order but not a Total Order
- C. A total Order
- D. An Equivalence Relation

gatecse-2006 set-theory&algebra normal relations

Answer key

4.12.28 Relations: GATE CSE 2007 | Question: 2



Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are:

- A. n and n B. n^2 and n C. n^2 and 0 D. n and 1

gatecse-2007 set-theory&algebra normal relations

Answer key

4.12.29 Relations: GATE CSE 2009 | Question: 4



Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. Which one of the following is **TRUE**?

- A. R is symmetric but NOT antisymmetric B. R is NOT symmetric but antisymmetric
C. R is both symmetric and antisymmetric D. R is neither symmetric nor antisymmetric

gatecse-2009 set-theory&algebra easy relations

Answer key

4.12.30 Relations: GATE CSE 2010 | Question: 3



What is the possible number of reflexive relations on a set of 5 elements?

- A. 2^{10} B. 2^{15} C. 2^{20} D. 2^{25}

gatecse-2010 set-theory&algebra easy relations

Answer key

4.12.31 Relations: GATE CSE 2015 | Set 2 | Question: 16



Let R be the relation on the set of positive integers such that aRb and only if a and b are distinct and let have a common divisor other than 1. Which one of the following statements about R is true?

- A. R is symmetric and reflexive but not transitive
B. R is reflexive but not symmetric not transitive
C. R is transitive but not reflexive and not symmetric
D. R is symmetric but not reflexive and not transitive

gatecse-2015-set2 set-theory&algebra relations normal

Answer key

4.12.32 Relations: GATE CSE 2015 | Set 3 | Question: 41



Let R be a relation on the set of ordered pairs of positive integers such that $((p, q), (r, s)) \in R$ if and only if $p - s = q - r$. Which one of the following is true about R ?

- A. Both reflexive and symmetric B. Reflexive but not symmetric
C. Not reflexive but symmetric D. Neither reflexive nor symmetric

gatecse-2015-set3 set-theory&algebra relations normal

Answer key

4.12.33 Relations: GATE CSE 2016 | Set 2 | Question: 26



A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a, b)R(c, d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

- $P : R$ is reflexive.
- $Q : R$ is transitive.

Which one of the following statements is **TRUE**?

- A. Both P and Q are true.
C. P is false and Q is true.

- B. P is true and Q is false.
D. Both P and Q are false.

gatecse-2016-set2 set-theory&algebra relations normal

Answer key

4.12.34 Relations: GATE CSE 2020 | Question: 17



Let $\mathcal{R}$ be the set of all binary relations on the set $\{1, 2, 3\}$. Suppose a relation is chosen from $\mathcal{R}$ at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

gatecse-2020 numerical-answers probability relations one-mark

Answer key

4.12.35 Relations: GATE CSE 2021 | Set 1 | Question: 43



A relation R is said to be circular if aRb and bRc together imply cRa .

Which of the following options is/are correct?

- A. If a relation S is reflexive and symmetric, then S is an equivalence relation.
B. If a relation S is circular and symmetric, then S is an equivalence relation.
C. If a relation S is reflexive and circular, then S is an equivalence relation.
D. If a relation S is transitive and circular, then S is an equivalence relation.

gatecse-2021-set1 multiple-selects set-theory&algebra relations two-marks

Answer key

4.12.36 Relations: GATE CSE 2025 | Set 2 | Question: 32



Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\preceq$ on $\mathcal{F}$ as follows:

$\forall f, g \in \mathcal{F}, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$.

Which of the following statement(s) is/are TRUE?

- A. $\preceq$ is a symmetric relation
B. $(\mathcal{F}, \preceq)$ is a partial order
C. $(\mathcal{F}, \preceq)$ is a lattice
D. $\preceq$ is an equivalence relation

gatecse2025-set2 set-theory&algebra functions relations multiple-selects two-marks

Answer key

4.12.37 Relations: GATE CSE 2026 | Set 2 | Question: 16



Let R be a binary relation on the set $\{1, 2, \dots, 10\}$, where $(x, y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R ?

- A. Reflexive B. Symmetric C. Transitive D. Antisymmetric

gatecse-2026-set2 set-theory&algebra relations multiple-selects one-mark

Answer key

4.12.38 Relations: GATE IT 2004 | Question: 4



Let R_1 be a relation from $A = \{1, 3, 5, 7\}$ to $B = \{2, 4, 6, 8\}$ and R_2 be another relation from B to $C = \{1, 2, 3, 4\}$ as defined below:

- i. An element x in A is related to an element y in B (under R_1) if $x + y$ is divisible by 3.
ii. An element x in B is related to an element y in C (under R_2) if $x + y$ is even but not divisible by 3.

Which is the composite relation $R_1 R_2$ from A to C ?

- A. $R_1 R_2 = \{(1, 2), (1, 4), (3, 3), (5, 4), (7, 3)\}$

- B. $R_1 R_2 = \{(1, 2), (1, 3), (3, 2), (5, 2), (7, 3)\}$
 C. $R_1 R_2 = \{(1, 2), (3, 2), (3, 4), (5, 4), (7, 2)\}$
 D. $R_1 R_2 = \{(3, 2), (3, 4), (5, 1), (5, 3), (7, 1)\}$

gateit-2004 set-theory&algebra relations normal

Answer key

4.13

Set Theory (27)

4.13.1 Set Theory: GATE CSE 1993 | Question: 17



Out of a group of 21 persons, 9 eat vegetables, 10 eat fish and 7 eat eggs. 5 persons eat all three. How many persons eat at least two out of the three dishes?

gate1993 set-theory&algebra easy set-theory descriptive

Answer key

4.13.2 Set Theory: GATE CSE 1993 | Question: 8.3



Let S be an infinite set and $S_1, \dots, S_n$ be sets such that $S_1 \cup S_2 \cup \dots \cup S_n = S$. Then

- A. at least one of the sets S_i is a finite set
 B. not more than one of the sets S_i can be finite
 C. at least one of the sets S_i is an infinite
 D. not more than one of the sets S_i can be infinite
 E. None of the above

gate1993 set-theory&algebra normal set-theory

Answer key

4.13.3 Set Theory: GATE CSE 1993 | Question: 8.4



Let A be a finite set of size n . The number of elements in the power set of $A \times A$ is:

- A. 2^{2n} B. 2^{n^2} C. $(2^n)^2$ D. $(2^2)^n$
 E. None of the above

gate1993 set-theory&algebra easy set-theory

Answer key

4.13.4 Set Theory: GATE CSE 1994 | Question: 2.4



The number of subsets $\{1, 2, \dots, n\}$ with odd cardinality is _____

gate1994 set-theory&algebra easy set-theory fill-in-the-blanks

Answer key

4.13.5 Set Theory: GATE CSE 1995 | Question: 1.20



The number of elements in the power set $P(S)$ of the set $S = \{\{\emptyset\}, 1, \{2, 3\}\}$ is:

- A. 2 B. 4 C. 8 D. None of the above

gate1995 set-theory&algebra normal set-theory

Answer key

4.13.6 Set Theory: GATE CSE 1995 | Question: 25b



Determine the number of positive integers (≤ 720) which are not divisible by any of 2, 3 or 5.

gate1995 set-theory&algebra set-theory numerical-answers

Answer key

4.13.7 Set Theory: GATE CSE 1996 | Question: 1.1



Let A and B be sets and let A^c and B^c denote the complements of the sets A and B . The set $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to

- A. $A \cup B$ B. $A^c \cup B^c$ C. $A \cap B$ D. $A^c \cap B^c$

gate1996 set-theory&algebra easy set-theory

Answer key

4.13.8 Set Theory: GATE CSE 1998 | Question: 2.4



In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada. 9 persons speak both English and Hindi, 11 persons speak both Hindi and Kannada whereas 13 persons speak both Kannada and English. How many speak all three languages?

- A. 9 B. 8 C. 7 D. 6

gate1998 set-theory&algebra easy set-theory

Answer key

4.13.9 Set Theory: GATE CSE 2000 | Question: 2.6



Let $P(S)$ denotes the power set of set S . Which of the following is always true?

- A. $P(P(S)) = P(S)$ B. $P(S) \cap P(P(S)) = \{\emptyset\}$
C. $P(S) \cap S = P(S)$ D. $S \notin P(S)$

gatecse-2000 set-theory&algebra easy set-theory

Answer key

4.13.10 Set Theory: GATE CSE 2000 | Question: 6



Let S be a set of n elements $\{1, 2, \dots, n\}$ and G a graph with 2^n vertices, each vertex corresponding to a distinct subset of S . Two vertices are adjacent iff the symmetric difference of the corresponding sets has exactly 2 elements. Note: The symmetric difference of two sets R_1 and R_2 is defined as $(R_1 \setminus R_2) \cup (R_2 \setminus R_1)$

Every vertex in G has the same degree. What is the degree of a vertex in G ?

How many connected components does G have?

gatecse-2000 set-theory&algebra normal descriptive set-theory

Answer key

4.13.11 Set Theory: GATE CSE 2001 | Question: 2.2



Consider the following statements:

- S_1 : There exists infinite sets A, B, C such that $A \cap (B \cup C)$ is finite.
- S_2 : There exists two irrational numbers x and y such that $(x + y)$ is rational.

Which of the following is true about S_1 and S_2 ?

- A. Only S_1 is correct B. Only S_2 is correct
C. Both S_1 and S_2 are correct D. None of S_1 and S_2 is correct

gatecse-2001 set-theory&algebra normal set-theory

Answer key

4.13.12 Set Theory: GATE CSE 2001 | Question: 3



a. Prove that $\text{powerset}(A \cap B) = \text{powerset}(A) \cap \text{powerset}(B)$

- b. Let $\text{sum}(n) = 0 + 1 + 2 + \dots + n$ for all natural numbers n . Give an induction proof to show that the following equation is true for all natural numbers m and n :

$$\text{sum}(m+n) = \text{sum}(m) + \text{sum}(n) + mn$$

gatecse-2001 set-theory&algebra normal set-theory descriptive

Answer key

4.13.13 Set Theory: GATE CSE 2005 | Question: 8



Let A, B and C be non-empty sets and let $X = (A - B) - C$ and $Y = (A - C) - (B - C)$. Which one of the following is TRUE?

- A. $X = Y$ B. $X \subset Y$ C. $Y \subset X$ D. None of these

gatecse-2005 set-theory&algebra easy set-theory

Answer key

4.13.14 Set Theory: GATE CSE 2006 | Question: 22



Let E, F and G be finite sets. Let

- $X = (E \cap F) - (F \cap G)$ and
- $Y = (E - (E \cap G)) - (E - F)$.

Which one of the following is true?

- A. $X \subset Y$ B. $X \supset Y$
C. $X = Y$ D. $X - Y \neq \emptyset$ and $Y - X \neq \emptyset$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.15 Set Theory: GATE CSE 2006 | Question: 24



Given a set of elements $N = 1, 2, \dots, n$ and two arbitrary subsets $A \subseteq N$ and $B \subseteq N$, how many of the $n!$ permutations π from N to N satisfy $\min(\pi(A)) = \min(\pi(B))$, where $\min(S)$ is the smallest integer in the set of integers S , and $\pi(S)$ is the set of integers obtained by applying permutation π to each element of S ?

- A. $(n - |A \cup B|)|A||B|$ B. $(|A|^2 + |B|^2)n^2$
C. $n! \frac{|A \cap B|}{|A \cup B|}$ D. $\frac{|A \cap B|^2}{n C_{|A \cup B|}}$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.16 Set Theory: GATE CSE 2008 | Question: 2



If P, Q, R are subsets of the universal set U , then

$$(P \cap Q \cap R) \cup (P^c \cap Q \cap R) \cup Q^c \cup R^c$$

is

- A. $Q^c \cup R^c$ B. $P \cup Q^c \cup R^c$
C. $P^c \cup Q^c \cup R^c$ D. U

gatecse-2008 normal set-theory&algebra set-theory

Answer key

4.13.17 Set Theory: GATE CSE 2014 | Set 2 | Question: 50



Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U .

Consider the following two statements:

- $S1$: There is a subset of S that is larger than every other subset.
- $S2$: There is a subset of S that is smaller than every other subset.

Which one of the following is CORRECT?

- A. Both $S1$ and $S2$ are true
- B. $S1$ is true and $S2$ is false
- C. $S2$ is true and $S1$ is false
- D. Neither $S1$ nor $S2$ is true

gatecse-2014-set2 set-theory&algebra normal set-theory

Answer key

4.13.18 Set Theory: GATE CSE 2015 | Set 1 | Question: 16



For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$, which of the following options are TRUE?

- I. $\emptyset \in 2^A$
- II. $\emptyset \subseteq 2^A$
- III. $\{5, \{6\}\} \in 2^A$
- IV. $\{5, \{6\}\} \subseteq 2^A$

- A. I and III only
- B. II and III only
- C. I, II and III only
- D. I, II and IV only

gatecse-2015-set1 set-theory&algebra set-theory normal

Answer key

4.13.19 Set Theory: GATE CSE 2015 | Set 2 | Question: 18



The cardinality of the power set of $\{0, 1, 2, \dots, 10\}$ is _____

gatecse-2015-set2 set-theory&algebra set-theory easy numerical-answers

Answer key

4.13.20 Set Theory: GATE CSE 2015 | Set 3 | Question: 23



Suppose U is the power set of the set $S = \{1, 2, 3, 4, 5, 6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T . For any $T, R \in U$ let $T \setminus R$ be the set of all elements in T which are not in R . Which one of the following is true?

- A. $\forall X \in U, (|X| = |X'|)$
- B. $\exists X \in U, \exists Y \in U, (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \emptyset)$
- C. $\forall X \in U, \forall Y \in U, (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \emptyset)$
- D. $\forall X \in U, \forall Y \in U, (X \setminus Y = Y' \setminus X')$

gatecse-2015-set3 set-theory&algebra set-theory normal

Answer key

4.13.21 Set Theory: GATE CSE 2016 | Set 2 | Question: 28



Consider a set U of 23 different compounds in a chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is ALWAYS TRUE?

- A. Only I
- B. Only II
- C. Only III
- D. None.

Answer key

4.13.22 Set Theory: GATE CSE 2017 | Set 1 | Question: 47



The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

Answer key

4.13.23 Set Theory: GATE CSE 2021 | Set 2 | Question: 37



For two n -dimensional real vectors P and Q , the operation $s(P, Q)$ is defined as follows:

$$s(P, Q) = \sum_{i=1}^n (P[i] \cdot Q[i])$$

Let $\mathcal{L}$ be a set of 10-dimensional non-zero real vectors such that for every pair of distinct vectors $P, Q \in \mathcal{L}$, $s(P, Q) = 0$. What is the maximum cardinality possible for the set $\mathcal{L}$?

- A. 9 B. 10 C. 11 D. 100

Answer key

4.13.24 Set Theory: GATE IT 2004 | Question: 2



In a class of 200 students, 125 students have taken Programming Language course, 85 students have taken Data Structures course, 65 students have taken Computer Organization course; 50 students have taken both Programming Language and Data Structures, 35 students have taken both Programming Language and Computer Organization; 30 students have taken both Data Structures and Computer Organization, 15 students have taken all the three courses.

How many students have not taken any of the three courses?

- A. 15 B. 20 C. 25 D. 30

Answer key

4.13.25 Set Theory: GATE IT 2005 | Question: 33



Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S_1 and S_2 in C , either $S_1 \subset S_2$ or $S_2 \subset S_1$. What is the maximum cardinality of C ?

- A. n B. $n + 1$ C. $2^{n-1} + 1$ D. $n!$

Answer key

4.13.26 Set Theory: GATE IT 2006 | Question: 23



Let P , Q and R be sets let Δ denote the symmetric difference operator defined as $P\Delta Q = (P \cup Q) - (P \cap Q)$. Using Venn diagrams, determine which of the following is/are TRUE?

- I. $P\Delta(Q \cap R) = (P\Delta Q) \cap (P\Delta R)$
 II. $P \cap (Q \cap R) = (P \cap Q) \Delta (P \Delta R)$

- A. I only B. II only C. Neither I nor II D. Both I and II

Answer key



What is the cardinality of the set of integers X defined below?

$$X = \{n \mid 1 \leq n \leq 123, n \text{ is not divisible by either } 2, 3 \text{ or } 5\}$$

A. 28

B. 33

C. 37

D. 44

gateit-2006 set-theory&algebra normal set-theory

Answer key

Answer Keys

4.1.1	N/A	4.1.2	N/A	4.1.3	C	4.1.4	D	4.1.5	A
4.1.6	A	4.1.7	B	4.1.8	A	4.2.1	True	4.2.2	D
4.3.1	N/A	4.3.2	N/A	4.3.3	N/A	4.3.4	B	4.3.5	A
4.3.6	C	4.3.7	N/A	4.3.8	C	4.3.9	A	4.3.10	N/A
4.3.11	A	4.3.12	B	4.3.13	B	4.3.14	D	4.3.15	B
4.3.16	C	4.3.17	C	4.3.18	16	4.3.19	D	4.3.20	B
4.3.21	A	4.3.22	36	4.3.23	0.95	4.3.24	2	4.3.25	B;C
4.3.26	B;C;D	4.3.27	2	4.3.28	10:10	4.3.29	C	4.3.30	D
4.4.1	N/A	4.4.2	N/A	4.4.3	N/A	4.4.4	N/A	4.4.5	C
4.4.6	D	4.4.7	N/A	4.4.8	C	4.4.9	C	4.4.10	D
4.4.11	N/A	4.4.12	N/A	4.4.13	N/A	4.4.14	B	4.4.15	A
4.4.16	D	4.4.17	C	4.4.18	A	4.4.19	C	4.4.20	A
4.4.21	A	4.4.22	C	4.4.23	A	4.4.24	5	4.4.25	4
4.4.26	42	4.4.27	B	4.4.28	7	4.4.29	B	4.4.30	A;B;C
4.4.31	A;D	4.4.32	B;D	4.4.33	4	4.5.1	B	4.6.1	N/A
4.6.2	N/A	4.6.3	D	4.6.4	5	4.6.5	B	4.6.6	N/A
4.6.7	B	4.6.8	D	4.6.9	0	4.6.10	A	4.7.1	N/A
4.7.2	N/A	4.8.1	24	4.8.2	8	4.8.3	36	4.8.4	D
4.8.5	D	4.8.6	C	4.8.7	N/A	4.9.1	D	4.10.1	N/A
4.10.2	E	4.10.3	B	4.10.4	A	4.10.5	N/A	4.10.6	D
4.10.7	A	4.10.8	C	4.10.9	5	4.10.10	D	4.11.1	B
4.11.2	D	4.11.3	A	4.11.4	8	4.12.1	False	4.12.2	N/A
4.12.3	N/A	4.12.4	N/A	4.12.5	N/A	4.12.6	N/A	4.12.7	D
4.12.8	B	4.12.9	N/A	4.12.10	N/A	4.12.11	A	4.12.12	B
4.12.13	C	4.12.14	N/A	4.12.15	N/A	4.12.16	B	4.12.17	C
4.12.18	A;C	4.12.19	N/A	4.12.20	C	4.12.21	B	4.12.22	D
4.12.23	N/A	4.12.24	B	4.12.25	C	4.12.26	D	4.12.27	A
4.12.28	B	4.12.29	D	4.12.30	C	4.12.31	D	4.12.32	C
4.12.33	B	4.12.34	0.1249:0.1259	4.12.35	C	4.12.36	B;C	4.12.37	A;B;C
4.12.38	C	4.13.1	N/A	4.13.2	C	4.13.3	B	4.13.4	N/A
4.13.5	C	4.13.6	192	4.13.7	A	4.13.8	D	4.13.9	B

4.13.10	N/A
4.13.15	C
4.13.20	D
4.13.25	B

4.13.11	C
4.13.16	D
4.13.21	B
4.13.26	C

4.13.12	N/A
4.13.17	A
4.13.22	271
4.13.27	B

4.13.13	A
4.13.18	C
4.13.23	B

4.13.14	C
4.13.19	2048
4.13.24	C